

CAREERS360
A Career is a Life

GATE
COMPUTER SCIENCE
& INFORMATION
2016-2018

QUESTION PAPER & ANSWER KEY

Q. 1 – Q. 5 carry one mark each.

Q.1 “From where are they bringing their books? _____ bringing _____ books from _____.”

The words that best fill the blanks in the above sentence are

- (A) Their, they’re, there
- (B) They’re, their, there
- (C) There, their, they’re
- (D) They’re, there, there

Q.2 “A _____ investigation can sometimes yield new facts, but typically organized ones are more successful.”

The word that best fills the blank in the above sentence is

- (A) meandering (B) timely (C) consistent (D) systematic

Q.3 The area of a square is d . What is the area of the circle which has the diagonal of the square as its diameter?

- (A) πd (B) πd^2 (C) $\frac{1}{4}\pi d^2$ (D) $\frac{1}{2}\pi d$

Q.4 What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

- (A) 3047 (B) 6047 (C) 7987 (D) 63847

Q.5 What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, _____, 0

- (A) 2880 (B) 1440 (C) 720 (D) 0

Q. 6 – Q. 10 carry two marks each.

- Q.6 In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen in the town and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $\frac{8}{9}$ th of the eligible men and $\frac{2}{3}$ rd of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

(A) 1,50,000 (B) 2,00,000
(C) 1,75,000 (D) 1,51,000

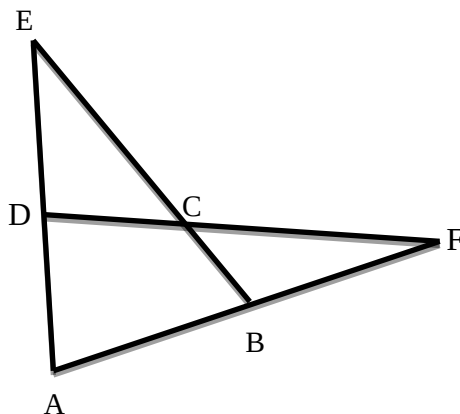
- Q.7 If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}$, $q^{-y} = \frac{1}{r}$, $r^{-z} = \frac{1}{p}$, what is the value of the product xyz ?

(A) -1 (B) $\frac{1}{pqr}$ (C) 1 (D) pqr

- Q.8 In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

(A) 2:3 (B) 1:1 (C) 3:2 (D) 2:1

- Q.9 In the figure below, $\angle DEC + \angle BFC$ is equal to _____.



(A) $\angle BCD - \angle BAD$ (B) $\angle BAD + \angle BCF$
(C) $\angle BAD + \angle BCD$ (D) $\angle CBA + \angle ADC$

- Q.10 A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?
- (A) Three green faces and four red faces.
 - (B) Four green faces and three red faces.
 - (C) Five green faces and two red faces.
 - (D) Six green faces and one red face.

END OF THE QUESTION PAPER

Q. 1 – Q. 25 carry one mark each.

Q.1 Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \dots$?

- (A) $\frac{3}{(1-x)^2}$ (B) $\frac{3x}{(1-x)^2}$ (C) $\frac{2-x}{(1-x)^2}$ (D) $\frac{3-x}{(1-x)^2}$

Q.2 Consider the following C program.

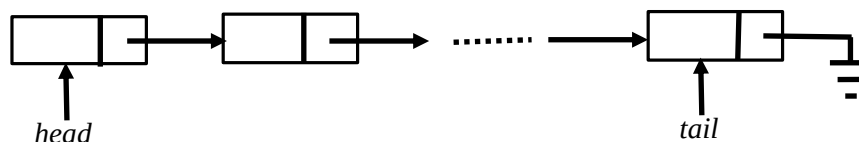
```
#include<stdio.h>
struct Ournode{
    char x,y,z;
};

int main(){
    struct Ournode p = {'1', '0', 'a'+2};
    struct Ournode *q = &p;
    printf ("%c, %c", *((char*)q+1), *((char*)q+2));
    return 0;
}
```

The output of this program is:

- (A) 0, c (B) 0, a+2 (C) '0', 'a+2' (D) '0', 'c'

Q.3 A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure?

- (A) $\theta(1), \theta(1)$ (B) $\theta(1), \theta(n)$ (C) $\theta(n), \theta(1)$ (D) $\theta(n), \theta(n)$

Q.4 Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

- (A) $\overline{P \oplus Q} = P \odot Q$
- (B) $\bar{P} \oplus Q = P \odot Q$
- (C) $\bar{P} \oplus \bar{Q} = P \oplus Q$
- (D) $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

Q.5 Consider the following processor design characteristics.

- I. Register-to-register arithmetic operations only
- II. Fixed-length instruction format
- III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

- (A) I and II only (B) II and III only (C) I and III only (D) I, II and III

Q.6 Let N be an NFA with n states. Let k be the number of states of a minimal DFA which is equivalent to N . Which one of the following is necessarily true?

- (A) $k \geq 2^n$ (B) $k \geq n$ (C) $k \leq n^2$ (D) $k \leq 2^n$

Q.7 The set of all recursively enumerable languages is

- (A) closed under complementation.
- (B) closed under intersection.
- (C) a subset of the set of all recursive languages.
- (D) an uncountable set.

Q.8 Which one of the following statements is FALSE?

- (A) Context-free grammar can be used to specify both lexical and syntax rules.
- (B) Type checking is done before parsing.
- (C) High-level language programs can be translated to different Intermediate Representations.
- (D) Arguments to a function can be passed using the program stack.

Q.9 The following are some events that occur after a device controller issues an interrupt while process L is under execution.

- (P) The processor pushes the process status of L onto the control stack.
- (Q) The processor finishes the execution of the current instruction.
- (R) The processor executes the interrupt service routine.
- (S) The processor pops the process status of L from the control stack.
- (T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur?

- (A) QPTRS (B) PTRSQ (C) TRPQS (D) QTPRS

Q.10 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

- (A) $(D - M) / (X - M)$ (B) $(X - M) / (D - M)$
(C) $(D - X) / (D - M)$ (D) $(X - M) / (D - X)$

Q.11 In an Entity-Relationship (ER) model, suppose R is a many-to-one relationship from entity set $E1$ to entity set $E2$. Assume that $E1$ and $E2$ participate totally in R and that the cardinality of $E1$ is greater than the cardinality of $E2$.

Which one of the following is true about R ?

- (A) Every entity in $E1$ is associated with exactly one entity in $E2$.
- (B) Some entity in $E1$ is associated with more than one entity in $E2$.
- (C) Every entity in $E2$ is associated with exactly one entity in $E1$.
- (D) Every entity in $E2$ is associated with at most one entity in $E1$.

Q.12 Consider the following two tables and four queries in SQL.

Book (isbn, bname), Stock (isbn, copies)

- Query 1: SELECT B.isbn, S.copies
 FROM Book B INNER JOIN Stock S
 ON B.isbn = S.isbn;
- Query 2: SELECT B.isbn, S.copies
 FROM Book B LEFT OUTER JOIN Stock S
 ON B.isbn = S.isbn;
- Query 3: SELECT B.isbn, S.copies
 FROM Book B RIGHT OUTER JOIN Stock S
 ON B.isbn = S.isbn;
- Query 4: SELECT B.isbn, S.copies
 FROM Book B FULL OUTER JOIN Stock S
 ON B.isbn = S.isbn;

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

- (A) Query 1 (B) Query 2 (C) Query 3 (D) Query 4

Q.13 Match the following:

<u>Field</u>	<u>Length in bits</u>
P. UDP Header's Port Number	I. 48
Q. Ethernet MAC Address	II. 8
R. IPv6 Next Header	III. 32
S. TCP Header's Sequence Number	IV. 16

- (A) P-III, Q-IV, R-II, S-I (B) P-II, Q-I, R-IV, S-III
(C) P-IV, Q-I, R-II, S-III (D) P-IV, Q-I, R-III, S-II

Q.14 Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that *cwnd* stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

- (i) The *cwnd* increases by 2 MSS on every successful acknowledgment.
- (ii) The *cwnd* approximately doubles on every successful acknowledgement.
- (iii) The *cwnd* increases by 1 MSS every round trip time.
- (iv) The *cwnd* approximately doubles every round trip time.

Which one of the following is correct?

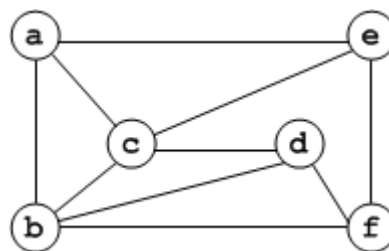
- (A) Only (ii) and (iii) are true
- (B) Only (i) and (iii) are true
- (C) Only (iv) is true
- (D) Only (i) and (iv) are true

Q.15 Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____.

Q.16 The value of $\int_0^{\pi/4} x \cos(x^2) dx$ correct to three decimal places (assuming that $\pi = 3.14$) is _____.

Q.17 Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v . The largest eigenvalue of A is _____.

Q.18 The chromatic number of the following graph is _____.



Q.19 Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

Q.20 The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

Q.21 Consider the following C program:

```
#include <stdio.h>

int counter = 0;

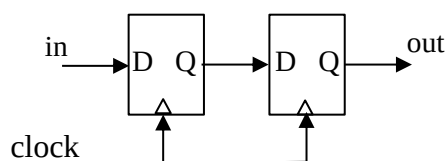
int calc (int a, int b) {
    int c;

    counter++;
    if (b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}

int main () {
    calc(4, 81);
    printf ("%d", counter);
}
```

The output of this program is _____.

Q.22 Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.



The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of “in” is _____.

- Q.23 A 32-bit wide main memory unit with a capacity of 1 GB is built using $256\text{M} \times 4\text{-bit}$ DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14} . The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is _____.
- Q.24 Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ____.
- Q.25 Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps ($= 10^9$ bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

Q. 26 – Q. 55 carry two marks each.

- Q.26 Consider a matrix $\mathbf{P}$ whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

Consider the following statements.

- (I) $\mathbf{P}$ does not have an inverse
- (II) $\mathbf{P}$ has a repeated eigenvalue
- (III) $\mathbf{P}$ cannot be diagonalized

Which one of the following options is correct?

- (A) Only I and III are necessarily true
- (B) Only II is necessarily true
- (C) Only I and II are necessarily true
- (D) Only II and III are necessarily true

Q.27 Let N be the set of natural numbers. Consider the following sets.

P : Set of Rational numbers (positive and negative)

Q : Set of functions from $\{0, 1\}$ to N

R : Set of functions from N to $\{0, 1\}$

S : Set of finite subsets of N .

Which of the sets above are countable?

- (A) Q and S only (B) P and S only (C) P and R only (D) P , Q and S only

Q.28 Consider the first-order logic sentence

$$\varphi \equiv \exists s \exists t \exists u \forall v \forall w \forall x \forall y \psi(s, t, u, v, w, x, y)$$

where $\psi(s, t, u, v, w, x, y)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

- (A) There exists at least one model of φ with universe of size less than or equal to 3.
(B) There exists no model of φ with universe of size less than or equal to 3.
(C) There exists no model of φ with universe of size greater than 7.
(D) Every model of φ has a universe of size equal to 7.

Q.29 Consider the following C program:

```
#include<stdio.h>

void fun1(char *s1, char *s2){
    char *tmp;
    tmp = s1;
    s1 = s2;
    s2 = tmp;
}

void fun2(char **s1, char **s2){
    char *tmp;
    tmp = *s1;
    *s1 = *s2;
    *s2 = tmp;
}

int main(){
    char *str1 = "Hi", *str2 = "Bye";
    fun1(str1, str2);    printf("%s %s ", str1, str2);
    fun2(&str1, &str2); printf("%s %s", str1, str2);
    return 0;
}
```

The output of the program above is

- | | |
|-------------------|-------------------|
| (A) Hi Bye Bye Hi | (B) Hi Bye Hi Bye |
| (C) Bye Hi Hi Bye | (D) Bye Hi Bye Hi |

Q.30 Let G be a simple undirected graph. Let T_D be a depth first search tree of G . Let T_B be a breadth first search tree of G . Consider the following statements.

- (I) No edge of G is a cross edge with respect to T_D . (A cross edge in G is between two nodes neither of which is an ancestor of the other in T_D .)
- (II) For every edge (u,v) of G , if u is at depth i and v is at depth j in T_B , then $|i - j| = 1$.

Which of the statements above must necessarily be true?

- | | |
|-------------------|----------------------|
| (A) I only | (B) II only |
| (C) Both I and II | (D) Neither I nor II |

- Q.31 Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1 G_2 G_3 \dots G_n$ can be done by parenthesizing in different ways. Define $G_i G_{i+1}$ as an **explicitly computed pair** for a given parenthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1 G_2 G_3 G_4 G_5 G_6$ using parenthesization $(G_1(G_2 G_3))(G_4(G_5 G_6))$, $G_2 G_3$ and $G_5 G_6$ are the only explicitly computed pairs.

Consider a matrix multiplication chain $F_1 F_2 F_3 F_4 F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are of dimensions $2 \times 25, 25 \times 3, 3 \times 16, 16 \times 1$ and 1×1000 , respectively. In the parenthesization of $F_1 F_2 F_3 F_4 F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

- (A) $F_1 F_2$ and $F_3 F_4$ only
 (B) $F_2 F_3$ only
 (C) $F_3 F_4$ only
 (D) $F_1 F_2$ and $F_4 F_5$ only
- Q.32 Consider the following C code. Assume that unsigned long int type length is 64 bits.

```
unsigned long int fun(unsigned long int n){
    unsigned long int i, j = 0, sum = 0;
    for (i = n; i > 1; i = i/2) j++;
    for ( ; j > 1; j = j/2) sum++;
    return(sum);
}
```

The value returned when we call fun with the input 2^{40} is

- (A) 4
 (B) 5
 (C) 6
 (D) 40
- Q.33 Consider the unsigned 8-bit fixed point binary number representation below,

$b_7 \ b_6 \ b_5 \ b_4 \ b_3 . b_2 \ b_1 \ b_0$

where the position of the binary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below **cannot** be represented **exactly** in the above representation:

(i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true?

- (A) None of (i), (ii), (iii), (iv) can be exactly represented
 (B) Only (ii) cannot be exactly represented
 (C) Only (iii) and (iv) cannot be exactly represented
 (D) Only (i) and (ii) cannot be exactly represented

Q.34 The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K -way set-associative cache memory, the length (in number of bits) of the tag field is

- (A) $P - N - \log_2 K$ (B) $P - N + \log_2 K$
 (C) $P - N - M - W - \log_2 K$ (D) $P - N - M - W + \log_2 K$

Q.35 Consider the following languages:

- I. $\{a^m b^n c^p d^q \mid m + p = n + q, \text{ where } m, n, p, q \geq 0\}$
 II. $\{a^m b^n c^p d^q \mid m = n \text{ and } p = q, \text{ where } m, n, p, q \geq 0\}$
 III. $\{a^m b^n c^p d^q \mid m = n = p \text{ and } p \neq q, \text{ where } m, n, p, q \geq 0\}$
 IV. $\{a^m b^n c^p d^q \mid mn = p + q, \text{ where } m, n, p, q \geq 0\}$

Which of the languages above are context-free?

- (A) I and IV only (B) I and II only
 (C) II and III only (D) II and IV only

Q.36 Consider the following problems. $L(G)$ denotes the language generated by a grammar G . $L(M)$ denotes the language accepted by a machine M .

- (I) For an unrestricted grammar G and a string w , whether $w \in L(G)$
 (II) Given a Turing machine M , whether $L(M)$ is regular
 (III) Given two grammars G_1 and G_2 , whether $L(G_1) = L(G_2)$
 (IV) Given an NFA N , whether there is a deterministic PDA P such that N and P accept the same language.

Which one of the following statements is correct?

- (A) Only I and II are undecidable (B) Only III is undecidable
 (C) Only II and IV are undecidable (D) Only I, II and III are undecidable

- Q.37 A lexical analyzer uses the following patterns to recognize three tokens T_1 , T_2 , and T_3 over the alphabet $\{a,b,c\}$.

$$T_1: a? (b|c)^* a$$

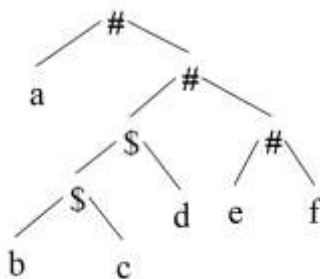
$$T_2: b? (a|c)^* b$$

$$T_3: c? (b|a)^* c$$

Note that 'x?' means 0 or 1 occurrence of the symbol x. Note also that the analyzer outputs the token that matches the longest possible prefix.

If the string *bbaacabc* is processed by the analyzer, which one of the following is the sequence of tokens it outputs?

- (A) $T_1 T_2 T_3$ (B) $T_1 T_1 T_3$ (C) $T_2 T_1 T_3$ (D) $T_3 T_3$
- Q.38 Consider the following parse tree for the expression $a\#b\$c\$d\#e\#f$, involving two binary operators $\$$ and $\#$.



Which one of the following is correct for the given parse tree?

- (A) $\$$ has higher precedence and is left associative; $\#$ is right associative
 (B) $\#$ has higher precedence and is left associative; $\$$ is right associative
 (C) $\$$ has higher precedence and is left associative; $\#$ is left associative
 (D) $\#$ has higher precedence and is right associative; $\$$ is left associative

- Q.39 In a system, there are three types of resources: E , F and G . Four processes P_0 , P_1 , P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, $\text{Max}[P_2, F]$ is the maximum number of instances of F that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available.

Allocation				Max			
	E	F	G		E	F	G
P_0	1	0	1	P_0	4	3	1
P_1	1	1	2	P_1	2	1	4
P_2	1	0	3	P_2	1	3	3
P_3	2	0	0	P_3	5	4	1

From the perspective of deadlock avoidance, which one of the following is true?

- (A) The system is in *safe* state.
- (B) The system is not in *safe* state, but would be *safe* if one more instance of E were available
- (C) The system is not in *safe* state, but would be *safe* if one more instance of F were available
- (D) The system is not in *safe* state, but would be *safe* if one more instance of G were available

- Q.40 Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N . Three semaphores *empty*, *full* and *mutex* are defined with respective initial values of 0, N and 1. Semaphore *empty* denotes the number of available slots in the buffer, for the consumer to read from. Semaphore *full* denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P , Q , R , and S , in the code below can be assigned either *empty* or *full*. The valid semaphore operations are: `wait()` and `signal()`.

Producer:	Consumer:
<pre>do{ wait(P); wait(mutex); //Add item to buffer signal(mutex); signal(Q); }while(1);</pre>	<pre>do{ wait(R); wait(mutex); //Consume item from buffer signal(mutex); signal(S); }while(1);</pre>

Which one of the following assignments to P , Q , R and S will yield the correct solution?

- (A) P : *full*, Q : *full*, R : *empty*, S : *empty*
 (B) P : *empty*, Q : *empty*, R : *full*, S : *full*
 (C) P : *full*, Q : *empty*, R : *empty*, S : *full*
 (D) P : *empty*, Q : *full*, R : *full*, S : *empty*
- Q.41 Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

$$Q: r \bowtie (\sigma_{B < 5}(s))$$

Let *LOJ* denote the natural left outer-join operation. Assume that r and s contain no null values.

Which one of the following queries is NOT equivalent to Q ?

- (A) $\sigma_{B < 5}(r \bowtie s)$ (B) $\sigma_{B < 5}(r \text{ LOJ } s)$
 (C) $r \text{ LOJ } (\sigma_{B < 5}(s))$ (D) $\sigma_{B < 5}(r) \text{ LOJ } s$

- Q.42 Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys.

Schema I: *Registration* (rollno, *courses*)

Field '*courses*' is a set-valued attribute containing the set of courses a student has registered for.

Non-trivial functional dependency:

$rollno \rightarrow courses$

Schema II: *Registration* (rollno, courseid, *email*)

Non-trivial functional dependencies:

$rollno, courseid \rightarrow email$

$email \rightarrow rollno$

Schema III: *Registration* (rollno, courseid, *marks*, *grade*)

Non-trivial functional dependencies:

$rollno, courseid \rightarrow marks, grade$

$marks \rightarrow grade$

Schema IV: *Registration* (rollno, courseid, *credit*)

Non-trivial functional dependencies:

$rollno, courseid \rightarrow credit$

$courseid \rightarrow credit$

Which one of the relational schemas above is in 3NF but not in BCNF?

- (A) Schema I (B) Schema II (C) Schema III (D) Schema IV

- Q.43 Let G be a graph with $100!$ vertices, with each vertex labelled by a distinct permutation of the numbers $1, 2, \dots, 100$. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v . Let y denote the degree of a vertex in G , and z denote the number of connected components in G . Then, $y + 10z = \underline{\hspace{2cm}}$.

- Q.44 Consider Guwahati (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D|H_G) = 0.40$. Similarly, the next two entries are $P(M_D|H_G) = 0.48$ and $P(L_D|H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$, and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

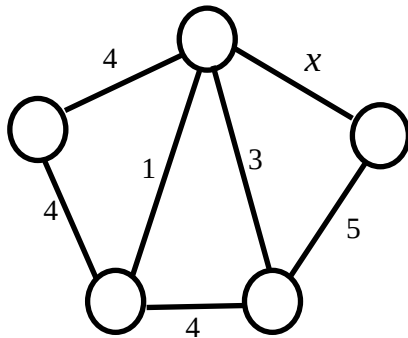
- Q.45 Consider the following program written in pseudo-code. Assume that x and y are integers.

```
Count(x, y) {
    if (y != 1) {
        if (x != 1) {
            print("*");
            Count(x/2, y);
        }
        else {
            y = y-1;
            Count(1024, y);
        }
    }
}
```

The number of times that the print statement is executed by the call $\text{Count}(1024, 1024)$ is _____.

- Q.46 The number of possible min-heaps containing each value from $\{1, 2, 3, 4, 5, 6, 7\}$ exactly once is _____.

Q.47 Consider the following undirected graph G :



Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G . The number of MWSTs of G for this value of x is _____.

Q.48 Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	Weight (in Kgs)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt} . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy} .

The value of $V_{\text{opt}} - V_{\text{greedy}}$ is _____.

Q.49 Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.

- Q.50 The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instructions is _____.

- Q.51 A processor has 16 integer registers (R0, R1, .. , R15) and 64 floating point registers (F0, F1, ... , F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F).

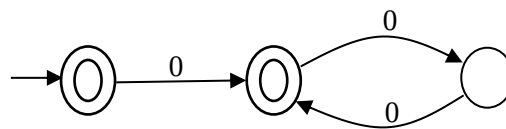
The maximum value of N is _____.

- Q.52 Given a language L , define L^i as follows:

$$L^0 = \{\epsilon\}$$

$$L^i = L^{i-1} \cdot L \text{ for all } i > 0$$

The order of a language L is defined as the smallest k such that $L^k = L^{k+1}$. Consider the language L_1 (over alphabet 0) accepted by the following automaton.



The order of L_1 is _____.

- Q.53 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

- Q.54 Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is _____.

- Q.55 Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance d meters from each other. P starts transmitting a packet at time $t=0$ after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time $t=0$ and begins to carrier-sense the medium.

The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is _____.

END OF THE QUESTION PAPER

Q.No.	Type	Section	Key/Range	Marks
1	MCQ	GA	B	1
2	MCQ	GA	A	1
3	MCQ	GA	D	1
4	MCQ	GA	C	1
5	MCQ	GA	B	1
6	MCQ	GA	B	2
7	MCQ	GA	C	2
8	MCQ	GA	B	2
9	MCQ	GA	A	2
10	MCQ	GA	C	2
1	MCQ	CS	D	1
2	MCQ	CS	A	1
3	MCQ	CS	B	1
4	MCQ	CS	D	1
5	MCQ	CS	D	1
6	MCQ	CS	D	1
7	MCQ	CS	B	1
8	MCQ	CS	B	1
9	MCQ	CS	A	1
10	MCQ	CS	B	1
11	MCQ	CS	A	1
12	MCQ	CS	D	1
13	MCQ	CS	C	1
14	MCQ	CS	C	1

Q.No.	Type	Section	Key/Range	Marks
15	NAT	CS	0.021 to 0.024	1
16	NAT	CS	0.27 to 0.30	1
17	NAT	CS	3 to 3	1
18	NAT	CS	3 to 3	1
19	NAT	CS	42 to 42	1
20	NAT	CS	4 to 4	1
21	NAT	CS	4 to 4	1
22	NAT	CS	2 to 2	1
23	NAT	CS	59.0 to 60.0	1
24	NAT	CS	2 to 2	1
25	NAT	CS	34 to 34	1
26	MCQ	CS	D	2
27	MCQ	CS	D	2
28	MCQ	CS	A	2
29	MCQ	CS	A	2
30	MCQ	CS	A	2
31	MCQ	CS	C	2
32	MCQ	CS	B	2
33	MCQ	CS	C	2
34	MCQ	CS	B	2
35	MCQ	CS	B	2
36	MCQ	CS	D	2
37	MCQ	CS	D	2
38	MCQ	CS	A	2

Q.No.	Type	Section	Key/Range	Marks
39	MCQ	CS	A	2
40	MCQ	CS	C	2
41	MCQ	CS	C	2
42	MCQ	CS	B	2
43	NAT	CS	109 to 109	2
44	NAT	CS	0.60 to 0.62	2
45	NAT	CS	10230 to 10230	2
46	NAT	CS	80 to 80	2
47	NAT	CS	4 to 4	2
48	NAT	CS	16 to 16	2
49	NAT	CS	3 to 3	2
50	NAT	CS	219 to 219	2
51	NAT	CS	32 to 32	2
52	NAT	CS	2 to 2	2
53	NAT	CS	85 to 85	2
54	NAT	CS	144 to 144	2
55	NAT	CS	50 to 50	2

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Question Paper Name:

Computer Science and Information Technology 11th Feb 2017 Session 1

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Duration:

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Organizing Institute:
Indian Institute of Technology Roorkee



Question Number : 1**Correct : 1 Wrong : -0.33**

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

- I. $p \Rightarrow q$
- II. $q \Rightarrow p$
- III. $(\neg q) \vee p$
- IV. $(\neg p) \vee q$

- (A) I only (B) I and IV only (C) II only (D) II and III only

Question Number : 2**Correct : 1 Wrong : -0.33**

Consider the first-order logic sentence $F: \forall x(\exists y R(x, y))$. Assuming non-empty logical domains, which of the sentences below are *implied* by F ?

- I. $\exists y(\exists x R(x, y))$
- II. $\exists y(\forall x R(x, y))$
- III. $\forall y(\exists x R(x, y))$
- IV. $\neg \exists x(\forall y \neg R(x, y))$

- (A) IV only (B) I and IV only (C) II only (D) II and III only

Question Number : 3**Correct : 1 Wrong : -0.33**

Let $c_1, \dots, c_n$ be scalars, not all zero, such that $\sum_{i=1}^n c_i a_i = 0$ where a_i are column vectors in $\mathbf{R}^n$.

Consider the set of linear equations

$$Ax = b$$

where $A = [a_1, \dots, a_n]$ and $b = \sum_{i=1}^n a_i$. The set of equations has

- (A) a unique solution at $x = J_n$ where J_n denotes a n -dimensional vector of all 1
- (B) no solution
- (C) infinitely many solutions
- (D) finitely many solutions

Question Number : 4**Correct : 1 Wrong : -0.33**

Consider the following functions from positive integers to real numbers:

$$10, \sqrt{n}, n, \log_2 n, \frac{100}{n}.$$

The CORRECT arrangement of the above functions in increasing order of asymptotic complexity is:

- (A) $\log_2 n, \frac{100}{n}, 10, \sqrt{n}, n$
- (B) $\frac{100}{n}, 10, \log_2 n, \sqrt{n}, n$
- (C) $10, \frac{100}{n}, \sqrt{n}, \log_2 n, n$
- (D) $\frac{100}{n}, \log_2 n, 10, \sqrt{n}, n$

Question Number : 5**Correct : 1 Wrong : -0.33**

Consider the following table:

Algorithms	Design Paradigms
(P) Kruskal	(i) Divide and Conquer
(Q) Quicksort	(ii) Greedy
(R) Floyd-Warshall	(iii) Dynamic Programming

Match the algorithms to the design paradigms they are based on.

- (A) $(P) \leftrightarrow (ii), (Q) \leftrightarrow (iii), (R) \leftrightarrow (i)$
(B) $(P) \leftrightarrow (iii), (Q) \leftrightarrow (i), (R) \leftrightarrow (ii)$
(C) $(P) \leftrightarrow (ii), (Q) \leftrightarrow (i), (R) \leftrightarrow (iii)$
(D) $(P) \leftrightarrow (i), (Q) \leftrightarrow (ii), (R) \leftrightarrow (iii)$

Question Number : 6**Correct : 1 Wrong : -0.33**

Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:
Note: The height of a tree with a single node is 0.

- (A) 4 and 15 respectively
(B) 3 and 14 respectively
(C) 4 and 14 respectively
(D) 3 and 15 respectively

Question Number : 7**Correct : 1 Wrong : -0.33**

The n -bit fixed-point representation of an unsigned real number X uses f bits for the fraction part. Let $i = n - f$. The range of decimal values for X in this representation is

- (A) 2^{-f} to 2^i (B) 2^{-f} to $(2^i - 2^{-f})$ (C) 0 to 2^i (D) 0 to $(2^i - 2^{-f})$

Question Number : 8**Correct : 1 Wrong : -0.33**

Consider the C code fragment given below.

```
typedef struct node {
    int data;
    node* next;
} node;

void join(node* m, node* n) {
    node* p = n;
    while (p->next != NULL) {
        p = p->next;
    }
    p->next = m;
}
```

Assuming that m and n point to valid NULL-terminated linked lists, invocation of join will

- (A) append list m to the end of list n for all inputs.
- (B) either cause a null pointer dereference or append list m to the end of list n.
- (C) cause a null pointer dereference for all inputs.
- (D) append list n to the end of list m for all inputs.

Question Number : 9**Correct : 1 Wrong : -0.33**

When two 8-bit numbers $A_7 \cdots A_0$ and $B_7 \cdots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a **ripple-carry adder**, the sum bits obtained are $S_7 \cdots S_0$ and the carry bits are $C_7 \cdots C_0$. An overflow is said to have occurred if

- (A) the carry bit C_7 is 1
- (B) all the carry bits $(C_7, \dots, C_0)$ are 1
- (C) $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1
- (D) $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1

Question Number : 10**Correct : 1 Wrong : -0.33**

Consider the following context-free grammar over the alphabet $\Sigma = \{a, b, c\}$ with S as the start symbol:

$$\begin{aligned} S &\rightarrow abScT \mid abcT \\ T &\rightarrow bT \mid b \end{aligned}$$

Which one of the following represents the language generated by the above grammar?

- (A) $\{(ab)^n(cb)^n \mid n \geq 1\}$
- (B) $\{(ab)^n cb^{m_1} cb^{m_2} \dots cb^{m_n} \mid n, m_1, m_2, \dots, m_n \geq 1\}$
- (C) $\{(ab)^n (cb^m)^n \mid m, n \geq 1\}$
- (D) $\{(ab)^n (cb^n)^m \mid m, n \geq 1\}$

Question Number : 11**Correct : 1 Wrong : -0.33**

Consider the C struct defined below:

```
struct data {  
    int marks [100];  
    char grade;  
    int cnumber;  
};  
struct data student;
```

The base address of `student` is available in register `R1`. The field `student.grade` can be accessed efficiently using

- (A) Post-increment addressing mode, $(R1)+$
- (B) Pre-decrement addressing mode, $-(R1)$
- (C) Register direct addressing mode, `R1`
- (D) Index addressing mode, $X(R1)$, where X is an offset represented in 2's complement 16-bit representation.

Consider the following intermediate program in three address code

$$\begin{aligned}p &= a - b \\q &= p * c \\p &= u * v \\q &= p + q\end{aligned}$$

Which one of the following corresponds to a *static single assignment* form of the above code?

(A)
$$\begin{aligned}p_1 &= a - b \\q_1 &= p_1 * c \\p_1 &= u * v \\q_1 &= p_1 + q_1\end{aligned}$$

(B)
$$\begin{aligned}p_3 &= a - b \\q_4 &= p_3 * c \\p_4 &= u * v \\q_5 &= p_4 + q_4\end{aligned}$$

(C)
$$\begin{aligned}p_1 &= a - b \\q_1 &= p_2 * c \\p_3 &= u * v \\q_2 &= p_4 + q_3\end{aligned}$$

(D)
$$\begin{aligned}p_1 &= a - b \\q_1 &= p * c \\p_2 &= u * v \\q_2 &= p + q\end{aligned}$$

Consider the following C code:

```
#include <stdio.h>
int *assignval(int *x, int val) {
    *x = val;
    return x;
}

void main () {
    int *x = malloc(sizeof(int));
    if(NULL == x) return;
    x = assignval(x,0);
    if(x) {
        x = (int *)malloc(sizeof(int));
        if(NULL == x) return;
        x = assignval(x,10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

- (A) compiler error as the return of malloc is not typecast appropriately
- (B) compiler error because the comparison should be made as `x == NULL` and not as shown
- (C) compiles successfully but execution may result in dangling pointer
- (D) compiles successfully but execution may result in memory leak

Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls `close` to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK, which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP?

- (A) LAST-ACK
- (B) TIME-WAIT
- (C) FIN-WAIT-1
- (D) FIN-WAIT-2

Question Number : 15**Correct : 1 Wrong : -0.33**

A sender S sends a message m to receiver R, which is digitally signed by S with its private key. In this scenario, one or more of the following security violations can take place.

- (I) S can launch a birthday attack to replace m with a fraudulent message.
- (II) A third party attacker can launch a birthday attack to replace m with a fraudulent message.
- (III) R can launch a birthday attack to replace m with a fraudulent message.

Which of the following are possible security violations?

- (A) (I) and (II) only
- (B) (I) only
- (C) (II) only
- (D) (II) and (III) only

Question Number : 16**Correct : 1 Wrong : -0.33**

The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

$$\begin{aligned}V &\rightarrow W \\ VW &\rightarrow X \\ Y &\rightarrow VX \\ Y &\rightarrow Z\end{aligned}$$

Which of the following is irreducible equivalent for this set of functional dependencies?

- | | | | | | | | |
|-----|-------------------|-----|-------------------|-----|-------------------|-----|-------------------|
| (A) | $V \rightarrow W$ | (B) | $V \rightarrow W$ | (C) | $V \rightarrow W$ | (D) | $V \rightarrow W$ |
| | $V \rightarrow X$ | | $W \rightarrow X$ | | $V \rightarrow X$ | | $W \rightarrow X$ |
| | $Y \rightarrow V$ | | $Y \rightarrow V$ | | $Y \rightarrow V$ | | $Y \rightarrow V$ |
| | $Y \rightarrow Z$ | | $Y \rightarrow Z$ | | $Y \rightarrow X$ | | $Y \rightarrow X$ |
| | | | | | $Y \rightarrow Z$ | | $Y \rightarrow Z$ |

Question Number : 17**Correct : 1 Wrong : -0.33**

Consider the following grammar:

$$\begin{array}{l} P \rightarrow xQRS \\ Q \rightarrow yz \mid z \\ R \rightarrow w \mid \varepsilon \\ S \rightarrow y \end{array}$$

What is FOLLOW(Q)?

- (A) $\{R\}$ (B) $\{w\}$ (C) $\{w, y\}$ (D) $\{w, \$\}$

Question Number : 18**Correct : 1 Wrong : -0.33**

Threads of a process share

- (A) global variables but not heap.
(B) heap but not global variables.
(C) neither global variables nor heap.
(D) both heap and global variables.

Question Number : 19**Correct : 1 Wrong : 0**

Let X be a Gaussian random variable with mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is _____.

Question Number : 20**Correct : 1 Wrong : 0**

Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is _____.

Question Number : 21**Correct : 1 Wrong : 0**

Consider the Karnaugh map given below, where X represents “don’t care” and blank represents 0.

$\begin{array}{c} \backslash \\ dc \end{array} \begin{array}{c} ba \end{array}$	00	01	11	10
00		x	x	
01	1			x
11	1			1
10		x	x	

Assume for all inputs (a, b, c, d) , the respective complements $(\bar{a}, \bar{b}, \bar{c}, \bar{d})$ are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

Question Number : 22**Correct : 1 Wrong : 0**

Consider the language L given by the regular expression $(a+b)^*b(a+b)$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is _____.

Question Number : 23**Correct : 1 Wrong : 0**

Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below.

EMP		
EmpId	EmpName	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
      (SELECT DeptName, COUNT(EmpId) AS
        EC(DeptName, Num)
       FROM EMP
        GROUP BY DeptName)
```

The output of executing the SQL query is _____.

Question Number : 24**Correct : 1 Wrong : 0**

Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) as given below :

Process	Arrival time	Burst time
P1	0	7
P2	3	3
P3	5	5
P4	6	2

If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then the average waiting time across all processes is _____ milliseconds.

Question Number : 25**Correct : 1 Wrong : 0**

Consider a two-level cache hierarchy with L1 and L2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L1 cache is 0.1; the L2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____.

Question Number : 26**Correct : 2 Wrong : -0.66**

Let $G = (V, E)$ be any connected undirected edge-weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements:

- (I) Minimum Spanning Tree of G is always unique.
- (II) Shortest path between any two vertices of G is always unique.

Which of the above statements is/are necessarily true?

- (A) (I) only
- (B) (II) only
- (C) both (I) and (II)
- (D) neither (I) nor (II)

Question Number : 27**Correct : 2 Wrong : -0.66**

A multithreaded program P executes with x number of threads and uses y number of locks for ensuring mutual exclusion while operating on shared memory locations. All locks in the program are *non-reentrant*, i.e., if a thread holds a lock l , then it cannot re-acquire lock l without releasing it. If a thread is unable to acquire a lock, it blocks until the lock becomes available. The *minimum* value of x and the *minimum* value of y together for which execution of P can result in a deadlock are:

- | | |
|--------------------|--------------------|
| (A) $x = 1, y = 2$ | (B) $x = 2, y = 1$ |
| (C) $x = 2, y = 2$ | (D) $x = 1, y = 1$ |

Question Number : 28**Correct : 2 Wrong : -0.66**

The value of $\lim_{x \rightarrow 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

- (A) is 0
- (B) is -1
- (C) is 1
- (D) does not exist

Question Number : 29**Correct : 2 Wrong : -0.66**

Let p , q , and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is

- (A) a tautology.
- (B) a contradiction.
- (C) always TRUE when p is FALSE.
- (D) always TRUE when q is TRUE.

Question Number : 30**Correct : 2 Wrong : -0.66**

Let u and v be two vectors in $\mathbf{R}^2$ whose Euclidean norms satisfy $\|u\| = 2\|v\|$. What is the value of α such that $w = u + \alpha v$ bisects the angle between u and v ?

- (A) 2
- (B) 1/2
- (C) 1
- (D) -1/2

Question Number : 31**Correct : 2 Wrong : -0.66**

Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2 = 50$. Consider the following statements.

- (I) One eigenvalue must be in $[-5, 5]$
- (II) The eigenvalue with the largest magnitude must be strictly greater than 5

Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

- (A) Both (I) and (II)
- (B) (I) only
- (C) (II) only
- (D) Neither (I) nor (II)

Question Number : 32**Correct : 2 Wrong : -0.66**

A computer network uses polynomials over $GF(2)$ for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as

(A) 01011011010

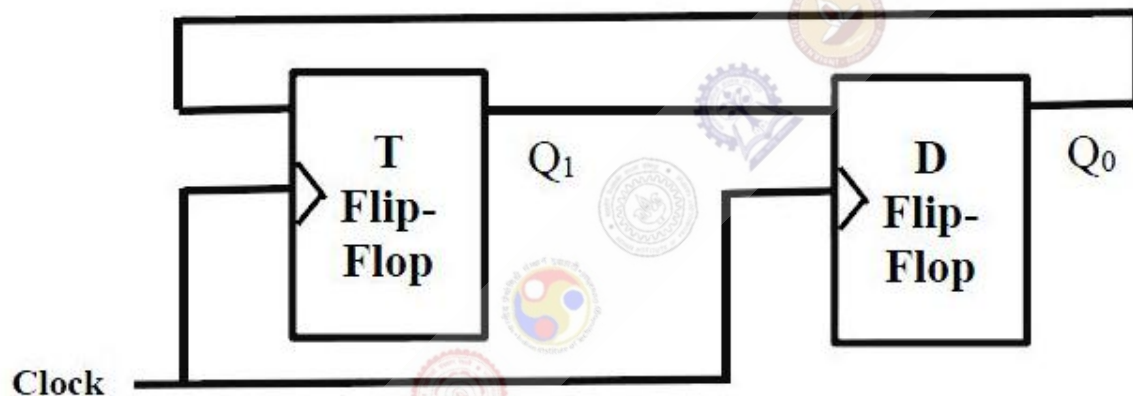
(B) 01011011011

(C) 01011011101

(D) 01011011100

Question Number : 33**Correct : 2 Wrong : -0.66**

Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

- (A) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 00 respectively
- (B) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 01 respectively
- (C) $Q_1 Q_0$ after the 3rd cycle are 00 and after the 4th cycle are 11 respectively
- (D) $Q_1 Q_0$ after the 3rd cycle are 01 and after the 4th cycle are 01 respectively

Question Number : 34**Correct : 2 Wrong : -0.66**

If G is a grammar with productions

$$S \rightarrow SaS \mid aSb \mid bSa \mid SS \mid \epsilon$$

where S is the start variable, then which one of the following strings is not generated by G ?

(A) $abab$ (B) $aaab$ (C) $abbaa$ (D) $babba$

Question Number : 35**Correct : 2 Wrong : -0.66**

Consider the following two functions.

```
void fun1(int n) {  
    if(n == 0) return;  
    printf("%d", n);  
    fun2(n - 2);  
    printf("%d", n);  
}
```

```
void fun2(int n) {  
    if(n == 0) return;  
    printf("%d", n);  
    fun1(++n);  
    printf("%d", n);  
}
```

The output printed when fun1 (5) is called is

- (A) 53423122233445
(C) 53423122132435

- (B) 53423120112233
(D) 53423120213243

Question Number : 36**Correct : 2 Wrong : -0.66**

Consider the C functions foo and bar given below:

```
int foo(int val) {  
    int x = 0;  
    while(val > 0) {  
        x = x + foo(val--);  
    }  
    return val;  
}
```

```
int bar(int val) {  
    int x = 0;  
    while(val > 0) {  
        x = x + bar(val-1);  
    }  
    return val;  
}
```

Invocations of foo (3) and bar (3) will result in :

- (A) Return of 6 and 6 respectively.
(B) Infinite loop and abnormal termination respectively.
(C) Abnormal termination and infinite loop respectively.
(D) Both terminating abnormally.

Question Number : 37**Correct : 2 Wrong : -0.66**

Consider the context-free grammars over the alphabet $\{a, b, c\}$ given below. S and T are non-terminals.

$$G_1: S \rightarrow aSb | T, T \rightarrow cT | \epsilon$$

$$G_2: S \rightarrow bSa | T, T \rightarrow cT | \epsilon$$

The language $L(G_1) \cap L(G_2)$ is

- (A) Finite. (B) Not finite but regular.
(C) Context-Free but not regular. (D) Recursive but not context-free.

Question Number : 38**Correct : 2 Wrong : -0.66**

Consider the following languages over the alphabet $\Sigma = \{a, b, c\}$.
Let $L_1 = \{a^n b^n c^m \mid m, n \geq 0\}$ and $L_2 = \{a^m b^n c^n \mid m, n \geq 0\}$.

Which of the following are context-free languages?

- I. $L_1 \cup L_2$
II. $L_1 \cap L_2$

- (A) I only (B) II only
(C) I and II (D) Neither I nor II

Question Number : 39**Correct : 2 Wrong : -0.66**

Let A and B be finite alphabets and let $\#$ be a symbol outside both A and B . Let f be a total function from A^* to B^* . We say f is *computable* if there exists a Turing machine M which given an input x in A^* , always halts with $f(x)$ on its tape. Let L_f denote the language $\{x\#f(x) \mid x \in A^*\}$. Which of the following statements is true:

- (A) f is computable if and only if L_f is recursive.
(B) f is computable if and only if L_f is recursively enumerable.
(C) If f is computable then L_f is recursive, but not conversely.
(D) If f is computable then L_f is recursively enumerable, but not conversely.

Question Number : 40**Correct : 2 Wrong : -0.66**

Recall that Belady's anomaly is that the page-fault rate may *increase* as the number of allocated frames increases. Now, consider the following statements:

S1: *Random page replacement* algorithm (where a page chosen at random is replaced) suffers from Belady's anomaly

S2: *LRU page replacement* algorithm suffers from Belady's anomaly

Which of the following is CORRECT?

(A) S1 is true, S2 is true

(B) S1 is true, S2 is false

(C) S1 is false, S2 is true

(D) S1 is false, S2 is false

Question Number : 41**Correct : 2 Wrong : -0.66**

Consider a database that has the relation schemas EMP(EmpId, EmpName, DeptId), and DEPT(DeptName, DeptId). Note that the DeptId can be permitted to be NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

(I) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \forall v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$

(II) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$

(III) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] = v[\text{DeptId}]))\}$

Which of the above queries are safe?

(A) (I) and (II) only

(B) (I) and (III) only

(C) (II) and (III) only

(D) (I), (II) and (III)

Question Number : 42**Correct : 2 Wrong : -0.66**

In a database system, unique timestamps are assigned to each transaction using Lamport's logical clock. Let $TS(T_1)$ and $TS(T_2)$ be the timestamps of transactions T_1 and T_2 respectively. Besides, T_1 holds a lock on the resource R , and T_2 has requested a conflicting lock on the same resource R . The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp.

```
if  $TS(T_2) < TS(T_1)$  then
     $T_1$  is killed
else  $T_2$  waits.
```

Assume any transaction that is not killed terminates eventually. Which of the following is TRUE about the database system that uses the above algorithm to prevent deadlocks?

- (A) The database system is both deadlock-free and starvation-free.
- (B) The database system is deadlock-free, but not starvation-free.
- (C) The database system is starvation-free, but not deadlock-free.
- (D) The database system is neither deadlock-free nor starvation-free.

Question Number : 43**Correct : 2 Wrong : 0**

Consider the following grammar:

```
stmt  -> if expr then expr else expr; stmt |  
expr  -> term relop term | term
term  -> id | number
id    -> a | b | c
number -> [0-9]
```

where **relop** is a relational operator (e.g., $<$, $>$, ...), ϵ refers to the empty statement, and **if**, **then**, **else** are terminals.

Consider a program P following the above grammar containing ten **if** terminals. The number of control flow paths in P is _____. For example, the program

if e_1 **then** e_2 **else** e_3

has 2 control flow paths, $e_1 \rightarrow e_2$ and $e_1 \rightarrow e_3$.

Question Number : 44**Correct : 2 Wrong : 0**

In a RSA cryptosystem, a participant A uses two prime numbers $p = 13$ and $q = 17$ to generate her public and private keys. If the public key of A is 35, then the private key of A is _____.

Question Number : 45**Correct : 2 Wrong : 0**

The values of parameters for the Stop-and-Wait ARQ protocol are as given below:

- Bit rate of the transmission channel = 1 Mbps.
- Propagation delay from sender to receiver = 0.75 ms.
- Time to process a frame = 0.25 ms.
- Number of bytes in the information frame = 1980.
- Number of bytes in the acknowledge frame = 20.
- Number of overhead bytes in the information frame = 20.

Assume that there are no transmission errors. Then, the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal places).

Question Number : 46**Correct : 2 Wrong : 0**

Consider a database that has the relation schema CR(StudentName, CourseName). An instance of the schema CR is as given below.

CR	
StudentName	CourseName
SA	CA
SA	CB
SA	CC
SB	CB
SB	CC
SC	CA
SC	CB
SC	CC
SD	CA
SD	CB
SD	CC
SD	CD
SE	CD
SE	CA
SE	CB
SF	CA
SF	CB
SF	CC

The following query is made on the database.

$$T1 \leftarrow \pi_{CourseName}(\sigma_{StudentName='SA'}(CR))$$

$$T2 \leftarrow CR \div T1$$

The number of rows in $T2$ is _____.

Question Number : 47**Correct : 2 Wrong : 0**

The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is _____.

Question Number : 48**Correct : 2 Wrong : 0**

Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A . The *worst case* number of probes performed by an *optimal* algorithm is _____.

Question Number : 49**Correct : 2 Wrong : 0**

Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC-relative addressing mode with Offset specified in bytes to the target location of the branch instruction. Further the Offset is always with respect to the address of the next instruction in the program sequence. Consider the following instruction sequence

<u>Instr. No.</u>	<u>Instruction</u>
i :	add R2, R3, R4
i+1 :	sub R5, R6, R7
i+2 :	cmp R1, R9, R10
i+3 :	beq R1, Offset

If the target of the branch instruction is i, then the decimal value of the Offset is _____.

Question Number : 50**Correct : 2 Wrong : 0**

Instruction execution in a processor is divided into 5 stages, *Instruction Fetch* (IF), *Instruction Decode* (ID), *Operand Fetch* (OF), *Execute* (EX), and *Write Back* (WB). These stages take **5, 4, 20, 10, and 3 nanoseconds (ns)** respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of **2 ns**. Two pipelined implementations of the processor are contemplated:

- (i) a naive pipeline implementation (NP) with 5 stages and
- (ii) an efficient pipeline (EP) where the OF stage is divided into stages OF1 and OF2 with execution times of **12 ns** and **8 ns** respectively.

The speedup (correct to two decimal places) achieved by EP over NP in executing 20 independent instructions with no hazards is _____.

Question Number : 51**Correct : 2 Wrong : 0**

Consider a 2-way set associative cache with 256 blocks and uses LRU replacement. Initially the cache is empty. Conflict misses are those misses which occur due to contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks

(0, 128, 256, 128, 0, 128, 256, 128, 1, 129, 257, 129, 1, 129, 257, 129)

is repeated 10 times. The number of *conflict misses* experienced by the cache is _____.

Question Number : 52**Correct : 2 Wrong : 0**

Consider the expression $(a-1)*(((b+c)/3)+d)$. Let X be the minimum number of registers required by an *optimal* code generation (without any register spill) algorithm for a load/store architecture, in which (i) *only load and store instructions can have memory operands* and (ii) *arithmetic instructions can have only register or immediate operands*. The value of X is _____.

Question Number : 53**Correct : 2 Wrong : 0**

Consider the following C program.

```
#include <stdio.h>
#include <string.h>

void printlength(char *s, char *t) {
    unsigned int c = 0;
    int len = ((strlen(s) - strlen(t)) > c) ? strlen(s) : strlen(t);
    printf("%d\n", len);
}

void main() {
    char *x = "abc";
    char *y = "defgh";
    printlength(x, y);
}
```

Recall that `strlen` is defined in `string.h` as returning a value of type `size_t`, which is an unsigned int. The output of the program is _____.

Question Number : 54**Correct : 2 Wrong : 0**

A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____ bits.

Question Number : 55**Correct : 2 Wrong : 0**

The output of executing the following C program is _____ .

```
#include <stdio.h>

int total(int v) {
    static int count = 0;
    while(v) {
        count += v&1;
        v >>= 1;
    }
    return count;
}

void main() {
    static int x = 0;
    int i = 5;
    for(; i > 0; i--) {
        x = x + total(i);
    }
    printf("%d\n", x);
}
```

Question Number : 56**Correct : 1 Wrong : -0.33**

After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

- (A) was wishing (B) is wishing (C) wished (D) had wished

Question Number : 57**Correct : 1 Wrong : -0.33**

Research in the workplace reveals that people work for many reasons _____.

- (A) money beside (B) beside money (C) money besides (D) besides money

Question Number : 58**Correct : 1 Wrong : -0.33**

Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

- (A) Rahul and Murali (B) Srinivas and Arul
(C) Srinivas and Murali (D) Srinivas and Rahul

Question Number : 59**Correct : 1 Wrong : -0.33**

Find the smallest number y such that $y \times 162$ is a perfect cube.

- (A) 24 (B) 27 (C) 32 (D) 36

Question Number : 60**Correct : 1 Wrong : -0.33**

The probability that a k -digit number does NOT contain the digits 0, 5, or 9 is

- (A) 0.3^k (B) 0.6^k (C) 0.7^k (D) 0.9^k

Question Number : 61**Correct : 2 Wrong : -0.66**

“The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history.”

Which of the following statements best reflects the author's opinion?

- (A) Nationalists are highly imaginative.
- (B) History is viewed through the filter of nationalism.
- (C) Our colonial past never happened.
- (D) Nationalism has to be both adequately and properly imagined.

Question Number : 62**Correct : 2 Wrong : -0.66**

Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

- (A) 2
- (B) 3
- (C) 4
- (D) Cannot be determined

Question Number : 63**Correct : 2 Wrong : -0.66**

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to

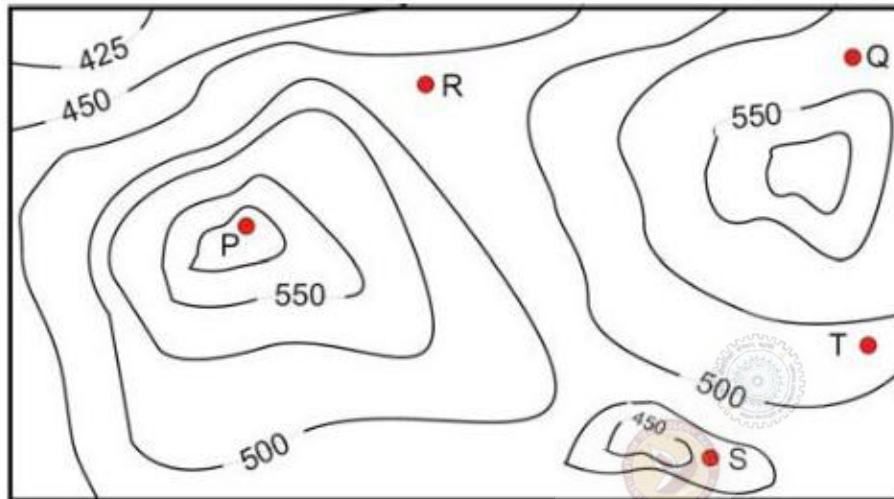
- (A) the maximum of x and y
- (B) the minimum of x and y
- (C) 1
- (D) none of the above

Question Number : 64**Correct : 2 Wrong : -0.66**

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

- (A) 21
- (B) 18
- (C) 16
- (D) 14

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?



(A) P, Q

(B) P, Q, T

(C) R, S, T

(D) Q, R, S

Q. No.	Type	Section	Key	Marks
1	MCQ	CS-1	D	1
2	MCQ	CS-1	B	1
3	MCQ	CS-1	C	1
4	MCQ	CS-1	B	1
5	MCQ	CS-1	C	1
6	MCQ	CS-1	B	1
7	MCQ	CS-1	D	1
8	MCQ	CS-1	B	1
9	MCQ	CS-1	C	1
10	MCQ	CS-1	B	1
11	MCQ	CS-1	D	1
12	MCQ	CS-1	B	1
13	MCQ	CS-1	D	1
14	MCQ	CS-1	D	1
15	MCQ	CS-1	B	1
16	MCQ	CS-1	A	1
17	MCQ	CS-1	C	1
18	MCQ	CS-1	D	1
19	NAT	CS-1	0.0 to 0.0	1
20	NAT	CS-1	18.0 to 18.0	1
21	NAT	CS-1	1.0 to 1.0	1
22	NAT	CS-1	4.0 to 4.0	1
23	NAT	CS-1	2.6 to 2.6	1
24	NAT	CS-1	3.0 to 3.0	1
25	NAT	CS-1	0.05 to 0.05	1
26	MCQ	CS-1	A	2
27	MCQ	CS-1	D	2
28	MCQ	CS-1	C	2
29	MCQ	CS-1	D	2
30	MCQ	CS-1	A	2
31	MCQ	CS-1	B	2
32	MCQ	CS-1	C	2
33	MCQ	CS-1	B	2
34	MCQ	CS-1	D	2
35	MCQ	CS-1	A	2
36	MCQ	CS-1	C	2

37	MCQ	CS-1	B	2
38	MCQ	CS-1	A	2
39	MCQ	CS-1	A	2
40	MCQ	CS-1	B	2
41	MCQ	CS-1	D	2
42	MCQ	CS-1	A	2
43	NAT	CS-1	1024.0 to 1024.0	2
44	NAT	CS-1	11.0 to 11.0	2
45	NAT	CS-1	86.5 to 89.5	2
46	NAT	CS-1	4.0 to 4.0	2
47	NAT	CS-1	271.0 to 271.0	2
48	NAT	CS-1	5.0 to 5.0	2
49	NAT	CS-1	-16.0 to -16.0	2
50	NAT	CS-1	1.49 to 1.52	2
51	NAT	CS-1	76.0 to 76.0	2
52	NAT	CS-1	2.0 to 2.0	2
53	NAT	CS-1	3.0 to 3.0	2
54	NAT	CS-1	14.0 to 14.0	2
55	NAT	CS-1	23.0 to 23.0	2
56	MCQ	GA	C	1
57	MCQ	GA	D	1
58	MCQ	GA	C	1
59	MCQ	GA	D	1
60	MCQ	GA	C	1
61	MCQ	GA	B	2
62	MCQ	GA	A	2
63	MCQ	GA	B	2
64	MCQ	GA	D	2
65	MCQ	GA	C	2

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Question Number : 1**Correct : 1 Wrong : -0.33**

The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is

- (A) 571244 (B) 736251 (C) 571247 (D) 136251

Question Number : 2**Correct : 1 Wrong : -0.33**

Match the following:

(P) <code>static char var;</code>	(i) Sequence of memory locations to store addresses
(Q) <code>m = malloc(10);</code> <code>m = NULL;</code>	(ii) A variable located in data section of memory
(R) <code>char *ptr[10];</code>	(iii) Request to allocate a CPU register to store data
(S) <code>register int var1;</code>	(iv) A lost memory which cannot be freed

- (A) $P \rightarrow (ii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (iii)$ (B) $P \rightarrow (ii), Q \rightarrow (i), R \rightarrow (iv), S \rightarrow (iii)$
 (C) $P \rightarrow (ii), Q \rightarrow (iv), R \rightarrow (iii), S \rightarrow (i)$ (D) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$

Question Number : 3**Correct : 1 Wrong : -0.33**

Match the algorithms with their time complexities:

<u>Algorithm</u>	<u>Time complexity</u>
(P) Towers of Hanoi with n disks	(i) $\Theta(n^2)$
(Q) Binary search given n sorted numbers	(ii) $\Theta(n \log n)$
(R) Heap sort given n numbers at the worst case	(iii) $\Theta(2^n)$
(S) Addition of two $n \times n$ matrices	(iv) $\Theta(\log n)$

- (A) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$
 (B) $P \rightarrow (iv), Q \rightarrow (iii), R \rightarrow (i), S \rightarrow (ii)$
 (C) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (ii), S \rightarrow (i)$
 (D) $P \rightarrow (iv), Q \rightarrow (iii), R \rightarrow (ii), S \rightarrow (i)$

Question Number : 4**Correct : 1 Wrong : -0.33**

Let L_1, L_2 be any two context-free languages and R be any regular language. Then which of the following is/are CORRECT?

- I. $L_1 \cup L_2$ is context-free.
- II. $\overline{L_1}$ is context-free.
- III. $L_1 - R$ is context-free.
- IV. $L_1 \cap L_2$ is context-free.

(A) I, II and IV only (B) I and III only (C) II and IV only (D) I only

Question Number : 5**Correct : 1 Wrong : -0.33**

Match the following according to input (from the left column) to the compiler phase (in the right column) that processes it:

(P) Syntax tree	(i) Code generator
(Q) Character stream	(ii) Syntax analyzer
(R) Intermediate representation	(iii) Semantic analyzer
(S) Token stream	(iv) Lexical analyzer

- (A) $P \rightarrow (ii), Q \rightarrow (iii), R \rightarrow (iv), S \rightarrow (i)$
- (B) $P \rightarrow (ii), Q \rightarrow (i), R \rightarrow (iii), S \rightarrow (iv)$
- (C) $P \rightarrow (iii), Q \rightarrow (iv), R \rightarrow (i), S \rightarrow (ii)$
- (D) $P \rightarrow (i), Q \rightarrow (iv), R \rightarrow (ii), S \rightarrow (iii)$

Question Number : 6**Correct : 1 Wrong : -0.33**

Which of the following statements about parser is/are CORRECT?

- I. Canonical LR is more powerful than SLR.
- II. SLR is more powerful than LALR.
- III. SLR is more powerful than Canonical LR.

(A) I only (B) II only (C) III only (D) II and III only

Question Number : 7

Correct : 1 Wrong : -0.33

Which of the following is/are shared by all the threads in a process?

- I. Program counter
- II. Stack
- III. Address space
- IV. Registers

(A) I and II only (B) III only (C) IV only (D) III and IV only

Question Number : 8

Correct : 1 Wrong : -0.33

In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed?

- I. Contiguous
- II. Linked
- III. Indexed

(A) I and III only (B) II only (C) III only (D) II and III only

Question Number : 9

Correct : 1 Wrong : -0.33

Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

- I: RIP uses distance vector routing
- II: RIP packets are sent using UDP
- III: OSPF packets are sent using TCP
- IV: OSPF operation is based on link-state routing

Which of the statements above are CORRECT?

- (A) I and IV only
- (B) I, II and III only
- (C) I, II and IV only
- (D) II, III and IV only

Question Number : 10**Correct : 1 Wrong : -0.33**

If $f(x) = R \sin\left(\frac{\pi x}{2}\right) + S$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x) dx = \frac{2R}{\pi}$, then the constants R and S are, respectively

- (A) $\frac{2}{\pi}$ and $\frac{16}{\pi}$ (B) $\frac{2}{\pi}$ and 0
(C) $\frac{4}{\pi}$ and 0 (D) $\frac{4}{\pi}$ and $\frac{16}{\pi}$

Question Number : 11**Correct : 1 Wrong : -0.33**

Let p, q, r denote the statements “It is raining”, “It is cold”, and “It is pleasant”, respectively. Then the statement “It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold” is represented by

- (A) $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$ (B) $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$
(C) $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$ (D) $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

Question Number : 12**Correct : 1 Wrong : -0.33**

Given the following binary number in 32-bit (single precision) IEEE-754 format:

00111111001101101000000000000000

The decimal value closest to this floating-point number is

- (A) 1.45×10^1 (B) 1.45×10^{-1} (C) 2.27×10^{-1} (D) 2.27×10^1

Question Number : 13**Correct : 1 Wrong : -0.33**

A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers **FRONT** and **REAR** pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in $O(1)$ time?

- I. Next pointer of front node points to the rear node.
II. Next pointer of rear node points to the front node.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Question Number : 14**Correct : 1 Wrong : -0.33**

Consider the following function implemented in C:

```
void printxy(int x, int y) {  
    int *ptr;  
    x = 0;  
    ptr = &x;  
    y = *ptr;  
    *ptr = 1;  
    printf("%d, %d", x, y);  
}
```

The output of invoking `printxy(1, 1)` is

(A) 0,0

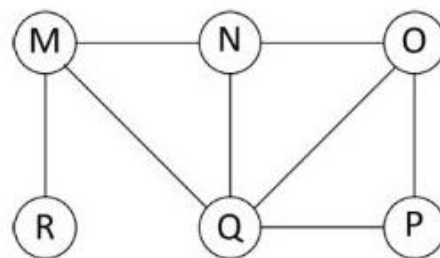
(B) 0,1

(C) 1,0

(D) 1,1

Question Number : 15**Correct : 1 Wrong : -0.33**

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



(A) MNOPQR

(B) NQMPOR

(C) QMNROP

(D) POQNMR

Question Number : 16**Correct : 1 Wrong : -0.33**

Identify the language generated by the following grammar, where S is the start variable.

$$\begin{aligned} S &\rightarrow XY \\ X &\rightarrow aX \mid a \\ Y &\rightarrow aYb \mid \epsilon \end{aligned}$$

- (A) $\{a^m b^n \mid m \geq n, n > 0\}$
(C) $\{a^m b^n \mid m > n, n \geq 0\}$

- (B) $\{a^m b^n \mid m \geq n, n \geq 0\}$
(D) $\{a^m b^n \mid m > n, n > 0\}$

Question Number : 17**Correct : 1 Wrong : -0.33**

An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions, can the relational table for R be merged with that of A?

- (A) Relationship R is one-to-many and the participation of A in R is total.
(B) Relationship R is one-to-many and the participation of A in R is partial.
(C) Relationship R is many-to-one and the participation of A in R is total.
(D) Relationship R is many-to-one and the participation of A in R is partial.

Question Number : 18**Correct : 1 Wrong : -0.33**

Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which **connect** function has already been called. Which of the following statements is/are CORRECT?

- I. A connected UDP socket can be used to communicate with multiple peers simultaneously.
II. A process can successfully call **connect** function again for an already connected UDP socket.
- (A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Question Number : 19**Correct : 1 Wrong : 0**

Consider the following tables T1 and T2.

T1		T2	
P	Q	R	S
2	2	2	2
3	8	8	3
7	3	3	2
5	8	9	7
6	9	5	7
8	5	7	2
9	8		

In table T1, **P** is the primary key and **Q** is the foreign key referencing **R** in table T2 with on-delete cascade and on-update cascade. In table T2, **R** is the primary key and **S** is the foreign key referencing **P** in table T1 with on-delete set NULL and on-update cascade. In order to delete record (3,8) from table T1, the number of additional records that need to be deleted from table T1 is _____.

Question Number : 20**Correct : 1 Wrong : 0**

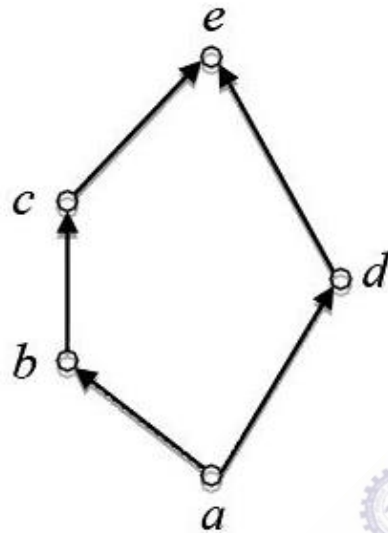
The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is _____.

Question Number : 21**Correct : 1 Wrong : 0**

Consider the set $X = \{a, b, c, d, e\}$ under the partial ordering

$$R = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, b), (b, c), (b, e), (c, c), (c, e), (d, d), (d, e), (e, e)\}.$$

The Hasse diagram of the partial order (X, R) is shown below.



The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is _____.

Question Number : 22**Correct : 1 Wrong : 0**

Let $P = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ be two matrices.

Then the rank of $P + Q$ is _____.

Question Number : 23**Correct : 1 Wrong : 0**

G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

Question Number : 24**Correct : 1 Wrong : 0**

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b . The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ _____.

Question Number : 25**Correct : 1 Wrong : 0**

The minimum possible number of states of a deterministic finite automaton that accepts the regular language $L = \{w_1aw_2 \mid w_1, w_2 \in \{a, b\}^*, |w_1| = 2, |w_2| \geq 3\}$ is _____.

Question Number : 26**Correct : 2 Wrong : -0.66**

P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for the job is

- (A) $\frac{4}{5}$ (B) $\frac{5}{6}$ (C) $\frac{7}{8}$ (D) $\frac{11}{12}$

Question Number : 27**Correct : 2 Wrong : -0.66**

If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

- (A) $wx + w(x + y) + x(x + y) = x + wy$
(B) $w\bar{x}(y + \bar{z}) + \bar{w}x = \bar{w} + x + \bar{y}z$
(C) $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$
(D) $(w + y)(wxy + wyz) = wxy + wyz$

Question Number : 28**Correct : 2 Wrong : -0.66**

Given $f(w, x, y, z) = \sum_m(0,1,2,3,7,8,10) + \sum_d(5,6,11,15)$, where d represents the *don't-care* condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$?

(A) $f = (\bar{w} + \bar{z})(\bar{x} + z)$

(B) $f = (\bar{w} + z)(x + z)$

(C) $f = (w + z)(\bar{x} + z)$

(D) $f = (w + \bar{z})(\bar{x} + z)$

Question Number : 29**Correct : 2 Wrong : -0.66**

In a two-level cache system, the access times of L_1 and L_2 caches are 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2 . The average memory access time (AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are:

(A) 0.111 and 0.056

(B) 0.056 and 0.111

(C) 0.0892 and 0.1784

(D) 0.1784 and 0.0892

Question Number : 30**Correct : 2 Wrong : -0.66**

Consider the recurrence function

$$T(n) = \begin{cases} 2T(\sqrt{n}) + 1, & n > 2 \\ 2, & 0 < n \leq 2 \end{cases}$$

Then $T(n)$ in terms of Θ notation is

(A) $\Theta(\log \log n)$

(B) $\Theta(\log n)$

(C) $\Theta(\sqrt{n})$

(D) $\Theta(n)$

Question Number : 31**Correct : 2 Wrong : -0.66**

For any discrete random variable X , with probability mass function

$P(X = j) = p_j, p_j \geq 0, j \in \{0, \dots, N\}$, and $\sum_{j=0}^N p_j = 1$, define the polynomial function

$g_X(z) = \sum_{j=0}^N p_j z^j$. For a certain discrete random variable Y , there exists a scalar $\beta \in [0, 1]$ such

that $g_Y(z) = (1 - \beta + \beta z)^N$. The expectation of Y is

- (A) $N\beta(1 - \beta)$
- (B) $N\beta$
- (C) $N(1 - \beta)$
- (D) Not expressible in terms of N and β alone

Question Number : 32**Correct : 2 Wrong : -0.66**

Consider the following expression grammar G :

$$\begin{aligned} E &\rightarrow E - T \mid T \\ T &\rightarrow T + F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

Which of the following grammars is not left recursive, but is equivalent to G ?

(A)
$$\begin{aligned} E &\rightarrow E - T \mid T \\ T &\rightarrow T + F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

(B)
$$\begin{aligned} E &\rightarrow TE' \\ E' &\rightarrow -TE' \mid \epsilon \\ T &\rightarrow T + F \mid F \\ F &\rightarrow (E) \mid id \end{aligned}$$

(C)
$$\begin{aligned} E &\rightarrow TX \\ X &\rightarrow -TX \mid \epsilon \\ T &\rightarrow FY \\ Y &\rightarrow +FY \mid \epsilon \\ F &\rightarrow (E) \mid id \end{aligned}$$

(D)
$$\begin{aligned} E &\rightarrow TX \mid (TX) \\ X &\rightarrow -TX \mid +TX \mid \epsilon \\ T &\rightarrow id \end{aligned}$$

Question Number : 33**Correct : 2 Wrong : -0.66**

A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below:

Process	Current Allocation	Maximum Requirement
P1	3	7
P2	1	6
P3	3	5

Which of the following best describes current state of the system?

- (A) Safe, Deadlocked
- (B) Safe, Not Deadlocked
- (C) Not Safe, Deadlocked
- (D) Not Safe, Not Deadlocked

Question Number : 34**Correct : 2 Wrong : -0.66**

Consider a binary code that consists of only four valid codewords as given below:

00000, 01011, 10101, 11110

Let the minimum Hamming distance of the code be p and the maximum number of erroneous bits that can be corrected by the code be q . Then the values of p and q are

- (A) $p=3$ and $q=1$
- (B) $p=3$ and $q=2$
- (C) $p=4$ and $q=1$
- (D) $p=4$ and $q=2$

Question Number : 35**Correct : 2 Wrong : -0.66**

Consider two hosts X and Y , connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec. Host X sends a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds, respectively. Then the values of p and q are

- (A) $p=50$ and $q=100$
- (B) $p=50$ and $q=400$
- (C) $p=100$ and $q=50$
- (D) $p=400$ and $q=50$

Question Number : 36**Correct : 2 Wrong : -0.66**

The pre-order traversal of a binary search tree is given by 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20. Then the post-order traversal of this tree is:

- (A) 2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20 (B) 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12
(C) 7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12 (D) 7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12

Question Number : 37**Correct : 2 Wrong : -0.66**

Consider the C program fragment below which is meant to divide x by y using repeated subtractions. The variables x , y , q and r are all unsigned int.

```
while (r >= y) {  
    r = r - y;  
    q = q + 1;  
}
```

Which of the following conditions on the variables x , y , q and r before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition $x == (y * q + r)$?

- (A) $(q == r) \ \&\& \ (r == 0)$
(B) $(x > 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
(C) $(q == 0) \ \&\& \ (r == x) \ \&\& \ (y > 0)$
(D) $(q == 0) \ \&\& \ (y > 0)$

Question Number : 38**Correct : 2 Wrong : -0.66**

Consider the following C function.

```
int fun(int n) {  
    int i, j;  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j < n; j += i) {  
            printf(" %d %d", i, j);  
        }  
    }  
}
```

Time complexity of fun in terms of Θ notation is

- (A) $\Theta(n\sqrt{n})$ (B) $\Theta(n^2)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2 \log n)$

Question Number : 39**Correct : 2 Wrong : -0.66**

Let δ denote the transition function and $\hat{\delta}$ denote the extended transition function of the ϵ -NFA whose transition table is given below:

δ	ϵ	a	b
$\rightarrow q_0$	$\{q_2\}$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
q_2	$\{q_0\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_2\}$

Then $\hat{\delta}(q_2, aba)$ is

- (A) $\emptyset$ (B) $\{q_0, q_1, q_3\}$ (C) $\{q_0, q_1, q_2\}$ (D) $\{q_0, q_2, q_3\}$

Question Number : 40**Correct : 2 Wrong : -0.66**

Consider the following languages.

$$L_1 = \{a^p \mid p \text{ is a prime number}\}$$

$$L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$$

$$L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$$

$$L_4 = \{a^n b^n \mid n \geq 1\}$$

Which of the following are CORRECT?

- I. L_1 is context-free but not regular.
 II. L_2 is not context-free.
 III. L_3 is not context-free but recursive.
 IV. L_4 is deterministic context-free.

- (A) I, II and IV only (B) II and III only (C) I and IV only (D) III and IV only

Question Number : 41

Correct : 2 Wrong : -0.66

Let $L(R)$ be the language represented by regular expression R . Let $L(G)$ be the language generated by a context free grammar G . Let $L(M)$ be the language accepted by a Turing machine M . Which of the following decision problems are undecidable?

- I. Given a regular expression R and a string w , is $w \in L(R)$?
- II. Given a context-free grammar G , is $L(G) = \emptyset$?
- III. Given a context-free grammar G , is $L(G) = \Sigma^*$ for some alphabet Σ ?
- IV. Given a Turing machine M and a string w , is $w \in L(M)$?

(A) I and IV only (B) II and III only (C) II, III and IV only (D) III and IV only

Question Number : 42

Correct : 2 Wrong : -0.66

The next state table of a 2-bit saturating up-counter is given below.

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T_1 and T_0 are

- (A) $T_1 = Q_1 Q_0, \quad T_0 = \bar{Q}_1 \bar{Q}_0$
- (B) $T_1 = \bar{Q}_1 Q_0, \quad T_0 = \bar{Q}_1 + \bar{Q}_0$
- (C) $T_1 = Q_1 + Q_0, \quad T_0 = \bar{Q}_1 + \bar{Q}_0$
- (D) $T_1 = \bar{Q}_1 Q_0, \quad T_0 = Q_1 + Q_0$

Consider the following snippet of a C program. Assume that `swap (&x, &y)` exchanges the contents of `x` and `y`.

```
int main() {  
    int array[] = {3, 5, 1, 4, 6, 2};  
    int done = 0;  
    int i;  
  
    while (done == 0) {  
        done = 1;  
        for (i=0; i<=4; i++) {  
            if (array[i] < array[i+1]) {  
                swap(&array[i], &array[i+1]);  
                done = 0;  
            }  
        }  
        for (i=5; i>=1; i--) {  
            if (array[i] > array[i-1]) {  
                swap(&array[i], &array[i-1]);  
                done = 0;  
            }  
        }  
    }  
    printf("%d", array[3]);  
}
```

The output of the program is _____.

Question Number : 44**Correct : 2 Wrong : 0**

Two transactions T_1 and T_2 are given as

$$T_1: r_1(X)w_1(X)r_1(Y)w_1(Y)$$

$$T_2: r_2(Y)w_2(Y)r_2(Z)w_2(Z)$$

where $r_i(V)$ denotes a *read* operation by transaction T_i on a variable V and $w_i(V)$ denotes a *write* operation by transaction T_i on a variable V . The total number of conflict serializable schedules that can be formed by T_1 and T_2 is _____.

Question Number : 45**Correct : 2 Wrong : 0**

The read access times and the hit ratios for different caches in a memory hierarchy are as given below.

Cache	Read access time (in nanoseconds)	Hit ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred-word-first read policy and the write back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____.

Consider the following database table named *top_scorer*.

top_scorer

player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Müller	Germany	14
Fontaine	France	13
Pelé	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Müller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player FROM top_scorer AS ta
WHERE ta.goals >ALL (SELECT tb.goals
                     FROM top_scorer AS tb
                     WHERE tb.country = 'Spain')
AND ta.goals >ANY (SELECT tc.goals
                  FROM top_scorer AS tc
                  WHERE tc.country = 'Germany')
```

The number of tuples returned by the above SQL query is _____.

If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to _____.

Question Number : 48**Correct : 2 Wrong : 0**

If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(X + 2)^2]$ equals _____.

Question Number : 49**Correct : 2 Wrong : 0**

In a B+ tree, if the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then the maximum order of the B+ tree is _____.

Question Number : 50**Correct : 2 Wrong : 0**

A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table of probabilities for each of the characters is shown below:

Character	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is _____.

Question Number : 51**Correct : 2 Wrong : 0**

Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is _____.

Question Number : 52**Correct : 2 Wrong : 0**

If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____.

Question Number : 53**Correct : 2 Wrong : 0**

Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

Question Number : 54

Correct : 2 Wrong : 0

Consider the following C Program.

```
#include<stdio.h>
int main() {
    int m = 10;
    int n, n1;
    n = ++m;
    n1 = m++;
    n--;
    --n1;
    n -= n1;
    printf("%d", n);
    return 0;
}
```

The output of the program is _____.

Question Number : 55

Correct : 2 Wrong : 0

Consider the following C Program.

```
#include<stdio.h>
#include<string.h>
int main() {
    char* c = "GATECSIT2017";
    char* p = c;
    printf("%d", (int)strlen(c+2[p]-6[p]-1));
    return 0;
}
```

The output of the program is _____.

Question Number : 56

Correct : 1 Wrong : -0.33

Choose the option with words that are not synonyms.

(A) aversion, dislike

(B) luminous, radiant

(C) plunder, loot

(D) yielding, resistant

Question Number : 57

Correct : 1 Wrong : -0.33

Saturn is _____ to be seen on a clear night with the naked eye.

(A) enough bright

(B) bright enough

(C) as enough bright

(D) bright as enough

Question Number : 58

Correct : 1 Wrong : -0.33

There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

(A) V

(B) W

(C) X

(D) Y

Question Number : 59

Correct : 1 Wrong : -0.33

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

(A) 12

(B) 15

(C) 18

(D) 19

Question Number : 60**Correct : 1 Wrong : -0.33**

There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

- (A) $1/5$ (B) $7/30$ (C) $1/4$ (D) $4/15$

Question Number : 61**Correct : 2 Wrong : -0.66**

“We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to something seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms.”

The author's belief that ideology is not as important as literature is revealed by the word:

- (A) ‘culture’ (B) ‘seemingly’ (C) ‘urgent’ (D) ‘political’

Question Number : 62**Correct : 2 Wrong : -0.66**

There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

- (A) The box labelled ‘Apples’ (B) The box labelled ‘Apples and Oranges’
(C) The box labelled ‘Oranges’ (D) Cannot be determined

Question Number : 63**Correct : 2 Wrong : -0.66**

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

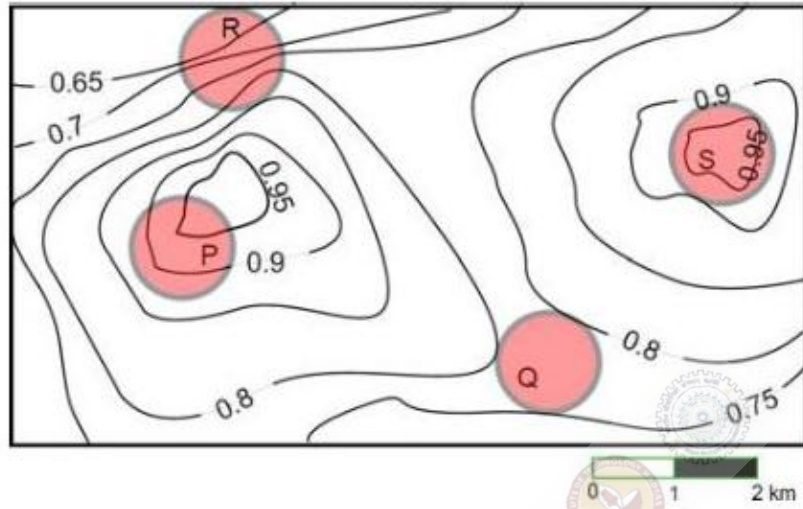
- (A) 90 digits (B) 91 digits (C) 92 digits (D) 93 digits

Question Number : 64**Correct : 2 Wrong : -0.66**

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

- (A) 0 (B) 1 (C) 2 (D) 3

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.



If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

- (A) P (B) Q (C) R (D) S

Q. No.	Type	Section	Key	Marks
1	MCQ	CS-2	D	1
2	MCQ	CS-2	A	1
3	MCQ	CS-2	C	1
4	MCQ	CS-2	B	1
5	MCQ	CS-2	C	1
6	MCQ	CS-2	A	1
7	MCQ	CS-2	B	1
8	MCQ	CS-2	D	1
9	MCQ	CS-2	C	1
10	MCQ	CS-2	C	1
11	MCQ	CS-2	A	1
12	MCQ	CS-2	C	1
13	MCQ	CS-2	B	1
14	MCQ	CS-2	C	1
15	MCQ	CS-2	D	1
16	MCQ	CS-2	C	1
17	MCQ	CS-2	C	1
18	MCQ	CS-2	B	1
19	NAT	CS-2	0.0 to 0.0	1
20	NAT	CS-2	9.0 to 9.0	1
21	NAT	CS-2	-0.01 to 0.01	1
22	NAT	CS-2	2.0 to 2.0	1
23	NAT	CS-2	16.0 to 16.0	1
24	NAT	CS-2	8.0 to 8.0	1
25	NAT	CS-2	8.0 to 8.0	1
26	MCQ	CS-2	A	2
27	MCQ	CS-2	C	2
28	MCQ	CS-2	A	2
29	MCQ	CS-2	A	2
30	MCQ	CS-2	B	2
31	MCQ	CS-2	B	2
32	MCQ	CS-2	C	2
33	MCQ	CS-2	B	2
34	MCQ	CS-2	A	2
35	MCQ	CS-2	D	2
36	MCQ	CS-2	B	2

37	MCQ	CS-2	C	2
38	MCQ	CS-2	C	2
39	MCQ	CS-2	C	2
40	MCQ	CS-2	D	2
41	MCQ	CS-2	D	2
42	MCQ	CS-2	B	2
43	NAT	CS-2	3.0 to 3.0	2
44	NAT	CS-2	54.0 to 54.0	2
45	NAT	CS-2	4.72 to 4.72	2
46	NAT	CS-2	7.0 to 7.0	2
47	NAT	CS-2	15.0 to 15.0	2
48	NAT	CS-2	54.0 to 54.0	2
49	NAT	CS-2	52.0 to 52.0	2
50	NAT	CS-2	225.0 to 225.0	2
51	NAT	CS-2	29.0 to 29.0	2
52	NAT	CS-2	5.0 to 5.0	2
53	NAT	CS-2	18.0 to 18.0	2
54	NAT	CS-2	0.0 to 0.0	2
55	NAT	CS-2	2.0 to 2.0	2
56	MCQ	GA	D	1
57	MCQ	GA	B	1
58	MCQ	GA	A	1
59	MCQ	GA	B	1
60	MCQ	GA	D	1
61	MCQ	GA	B	2
62	MCQ	GA	B	2
63	MCQ	GA	A	2
64	MCQ	GA	C	2
65	MCQ	GA	C	2

Q. 1 – Q. 5 carry one mark each.

Q.1 Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

- (A) I will not leave the place until the minister does not meet me.
- (B) I will not leave the place until the minister doesn't meet me.
- (C) I will not leave the place until the minister meet me.
- (D) I will not leave the place until the minister meets me.

Q.2 A rewording of something written or spoken is a _____.

- (A) paraphrase
- (B) paradox
- (C) paradigm
- (D) paraffin

Q.3 Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

- (A) figurative
- (B) collateral
- (C) literal
- (D) figurine

Q.4 If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

- (A) zentaga
- (B) tagafer
- (C) tagazen
- (D) relffer

Q.5 A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

- (A) 56
- (B) 64
- (C) 72
- (D) 96

Q. 6 – Q. 10 carry two marks each.

- Q.6 A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter \ Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

- (A) Elegance (B) Executive (C) Smooth (D) Soft

- Q.7 Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.

Which of the following can be logically inferred from the above sentences?

- (A) India is a country of exactly seventeen languages.
 (B) Linguistic pluralism is the only indicator of a nation's diversity.
 (C) Indian currency notes have sufficient space for all the Indian languages.
 (D) Linguistic pluralism is strong evidence of India's diversity.

- Q.8 Consider the following statements relating to the level of poker play of four players **P**, **Q**, **R** and **S**.

- I. **P** always beats **Q**
 II. **R** always beats **S**
 III. **S** loses to **P** only sometimes
 IV. **R** always loses to **Q**

Which of the following can be logically inferred from the above statements?

- (i) **P** is likely to beat all the three other players
 (ii) **S** is the absolute worst player in the set

- (A) (i) only (B) (ii) only (C) (i) and (ii) (D) neither (i) nor (ii)

- Q.9 If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

- (A) (x^3+8) (B) $(x-1)$ (C) $(2x-5)$ (D) $(x+1)$

Q.10 In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

(A) 40.00

(B) 46.02

(C) 60.01

(D) 92.02

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Let p, q, r, s represent the following propositions.

p : $x \in \{8, 9, 10, 11, 12\}$

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____ .

Q.2 Let a_n be the number of n -bit strings that do NOT contain two consecutive 1s. Which one of the following is the recurrence relation for a_n ?

(A) $a_n = a_{n-1} + 2a_{n-2}$

(B) $a_n = a_{n-1} + a_{n-2}$

(C) $a_n = 2a_{n-1} + a_{n-2}$

(D) $a_n = 2a_{n-1} + 2a_{n-2}$

Q.3

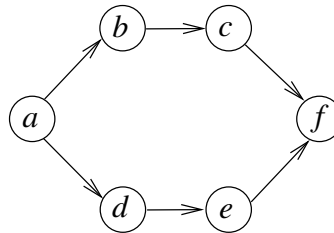
$$\lim_{x \rightarrow 4} \frac{\sin(x-4)}{x-4} = \text{_____}.$$

Q.4 A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is _____ .

Q.5 Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____ .

- Q.6** Consider the Boolean operator # with the following properties:
 $x \# 0 = x$, $x \# 1 = \bar{x}$, $x \# x = 0$ and $x \# \bar{x} = 1$. Then $x \# y$ is equivalent to
- (A) $x\bar{y} + \bar{x}y$
 - (B) $x\bar{y} + \bar{x}\bar{y}$
 - (C) $\bar{x}y + xy$
 - (D) $xy + \bar{x}\bar{y}$
- Q.7** The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is _____.
- Q.8** We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is _____.
- Q.9** A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least _____ bits.
- Q.10** A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is **CORRECT** (n refers to the number of items in the queue)?
- (A) Both operations can be performed in $O(1)$ time
 - (B) At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$
 - (C) The worst case time complexity for both operations will be $\Omega(n)$
 - (D) Worst case time complexity for both operations will be $\Omega(\log n)$

Q.11 Consider the following directed graph:



The number of different topological orderings of the vertices of the graph is _____ .

Q.12 Consider the following C program.

```
void f(int, short);  
void main()  
{  
    int i = 100;  
    short s = 12;  
    short *p = &s;  
    _____ ;    // call to f()  
}
```

Which one of the following expressions, when placed in the blank above, will **NOT** result in a type checking error?

- (A) `f(s, *s)`
- (B) `i = f(i, s)`
- (C) `f(i, *s)`
- (D) `f(i, *p)`

Q.13 The worst case running times of *Insertion sort*, *Merge sort* and *Quick sort*, respectively, are:

- (A) $\Theta(n \log n)$, $\Theta(n \log n)$, and $\Theta(n^2)$
- (B) $\Theta(n^2)$, $\Theta(n^2)$, and $\Theta(n \log n)$
- (C) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n \log n)$
- (D) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n^2)$

Q.14 Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are **TRUE**?

P: Minimum spanning tree of G does not change

Q: Shortest path between any pair of vertices does not change

- (A) P only
- (B) Q only
- (C) Neither P nor Q
- (D) Both P and Q

Q.15 Consider the following C program.

```
#include<stdio.h>
void mystery(int *ptrb, int *ptrb) {
    int *temp;
    temp = ptrb;
    ptrb = ptrb;
    ptrb = temp;
}
int main() {
    int a=2016, b=0, c=4, d=42;
    mystery(&a, &b);
    if (a < c)
        mystery(&c, &a);
    mystery(&a, &d);
    printf("%d\n", a);
}
```

The output of the program is _____ .

Q.16 Which of the following languages is generated by the given grammar?

$$S \longrightarrow aS \mid bS \mid \varepsilon$$

- (A) $\{a^n b^m \mid n, m \geq 0\}$
- (B) $\{w \in \{a, b\}^* \mid w \text{ has equal number of a's and b's}\}$
- (C) $\{a^n \mid n \geq 0\} \cup \{b^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$
- (D) $\{a, b\}^*$

Q.17 Which of the following decision problems are undecidable?

- I. Given NFAs N_1 and N_2 , is $L(N_1) \cap L(N_2) = \Phi$?
- II. Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$?
- III. Given CFGs G_1 and G_2 , is $L(G_1) = L(G_2)$?
- IV. Given a TM M , is $L(M) = \Phi$?

- (A) I and IV only
- (B) II and III only
- (C) III and IV only
- (D) II and IV only

Q.18 Which one of the following regular expressions represents the language: *the set of all binary strings having two consecutive 0s and two consecutive 1s*?

- (A) $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$
- (B) $(0+1)^*(00(0+1)^*11 + 11(0+1)^*00)(0+1)^*$
- (C) $(0+1)^*00(0+1)^* + (0+1)^*11(0+1)^*$
- (D) $00(0+1)^*11 + 11(0+1)^*00$

Q.19 Consider the following code segment.

```
x = u - t;  
y = x * v;  
x = y + w;  
y = t - z;  
y = x * y;
```

The minimum number of *total* variables required to convert the above code segment to *static single assignment* form is _____.

- Q.20** Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?
- (A) Shortest remaining time first
 - (B) Round-robin with time quantum less than the shortest CPU burst
 - (C) Uniform random
 - (D) Highest priority first with priority proportional to CPU burst length
- Q.21** Which of the following is **NOT** a superkey in a relational schema with attributes V, W, X, Y, Z and primary key VY ?
- (A) $VXYZ$
 - (B) $VWXZ$
 - (C) $VWXY$
 - (D) $VWXYZ$
- Q.22** Which one of the following is **NOT** a part of the ACID properties of database transactions?
- (A) Atomicity
 - (B) Consistency
 - (C) Isolation
 - (D) Deadlock-freedom

Q.23 A database of research articles in a journal uses the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)

The primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE) and the following functional dependencies exist in the schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ TITLE

(VOLUME, NUMBER) $\rightarrow$ YEAR

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ PRICE

The database is redesigned to use the following schemas.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE)

(VOLUME, NUMBER, YEAR)

Which is the weakest normal form that the new database satisfies, but the old one does not?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Q.24 Which one of the following protocols is **NOT** used to resolve one form of address to another one?

- (A) DNS
- (B) ARP
- (C) DHCP
- (D) RARP

Q.25 Which of the following is/are example(s) of stateful application layer protocols?

- (i) HTTP
- (ii) FTP
- (iii) TCP
- (iv) POP3

- (A) (i) and (ii) only
- (B) (ii) and (iii) only
- (C) (ii) and (iv) only
- (D) (iv) only

Q. 26 - Q. 55 carry two marks each.

Q.26 The coefficient of x^{12} in $(x^3 + x^4 + x^5 + x^6 + \dots)^3$ is _____ .

Q.27 Consider the recurrence relation $a_1 = 8$, $a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 10^4$. The value of K is _____ .

Q.28 A function $f : \mathbb{N}^+ \rightarrow \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$\begin{aligned} f(n) &= f(n/2) && \text{if } n \text{ is even} \\ f(n) &= f(n+5) && \text{if } n \text{ is odd} \end{aligned}$$

Let $R = \{i \mid \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____ .

Q.29 Consider the following experiment.

Step 1. Flip a fair coin twice.

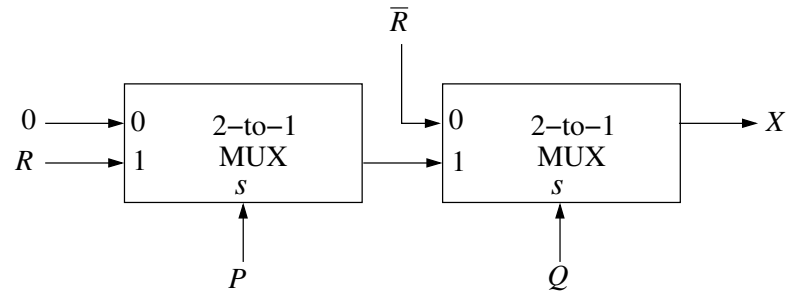
Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places) _____ .

Q.30 Consider the two cascaded 2-to-1 multiplexers as shown in the figure.



The minimal sum of products form of the output X is

- (A) $\bar{P}\bar{Q} + PQR$
- (B) $\bar{P}Q + QR$
- (C) $PQ + \bar{P}\bar{Q}R$
- (D) $\bar{Q}\bar{R} + PQR$

Q.31 The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is _____.

Q.32 The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is _____ percent.

Q.33 Consider a carry lookahead adder for adding two n -bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is

- (A) $\Theta(1)$
- (B) $\Theta(\log(n))$
- (C) $\Theta(\sqrt{n})$
- (D) $\Theta(n)$

Q.34 The following function computes the maximum value contained in an integer array `p[]` of size `n` (`n >= 1`).

```
int max(int *p, int n) {  
    int a=0, b=n-1;  
  
    while (_____) {  
        if (p[a] <= p[b]) { a = a+1; }  
        else                { b = b-1; }  
    }  
  
    return p[a];  
}
```

The missing loop condition is

- (A) `a != n`
- (B) `b != 0`
- (C) `b > (a + 1)`
- (D) `b != a`

Q.35 What will be the output of the following C program?

```
void count(int n){  
    static int d=1;  
  
    printf("%d ", n);  
    printf("%d ", d);  
    d++;  
    if(n>1) count(n-1);  
    printf("%d ", d);  
}
```

```
void main(){  
    count(3);  
}
```

- (A) 3 1 2 2 1 3 4 4 4
- (B) 3 1 2 1 1 1 2 2 2
- (C) 3 1 2 2 1 3 4
- (D) 3 1 2 1 1 1 2

- Q.36** What will be the output of the following pseudo-code when parameters are passed by reference and dynamic scoping is assumed?

```
a=3;
void n(x) {x = x * a; print(x);}
void m(y) {a = 1; a = y - a; n(a); print(a);}
void main() {m(a);}
```

- (A) 6, 2
(B) 6, 6
(C) 4, 2
(D) 4, 4
- Q.37** An operator `delete(i)` for a binary heap data structure is to be designed to delete the item in the *i*-th node. Assume that the heap is implemented in an array and *i* refers to the *i*-th index of the array. If the heap tree has depth *d* (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?
- (A) $O(1)$
(B) $O(d)$ but not $O(1)$
(C) $O(2^d)$ but not $O(d)$
(D) $O(d2^d)$ but not $O(2^d)$
- Q.38** Consider the weighted undirected graph with 4 vertices, where the weight of edge $\{i, j\}$ is given by the entry W_{ij} in the matrix *W*.

$$W = \begin{bmatrix} 0 & 2 & 8 & 5 \\ 2 & 0 & 5 & 8 \\ 8 & 5 & 0 & x \\ 5 & 8 & x & 0 \end{bmatrix}$$

The largest possible integer value of *x*, for which at least one shortest path between some pair of vertices will contain the edge with weight *x* is _____.

- Q.39** Let *G* be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of *G* can have is _____.

Q.40 $G = (V, E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G . Which of the following statements about the minimum spanning trees (MSTs) of G is/are **TRUE**?

- I. If e is the lightest edge of some cycle in G , then every MST of G includes e
- II. If e is the heaviest edge of some cycle in G , then every MST of G excludes e

(A) I only (B) II only (C) both I and II (D) neither I nor II

Q.41 Let Q denote a queue containing sixteen numbers and S be an empty stack. $\text{Head}(Q)$ returns the element at the head of the queue Q **without** removing it from Q . Similarly $\text{Top}(S)$ returns the element at the top of S **without** removing it from S . Consider the algorithm given below.

```

while  $Q$  is not Empty do
    if  $S$  is Empty OR  $\text{Top}(S) \leq \text{Head}(Q)$  then
         $x := \text{Dequeue}(Q)$ ;
         $\text{Push}(S, x)$ ;
    else
         $x := \text{Pop}(S)$ ;
         $\text{Enqueue}(Q, x)$ ;
    end
end

```

The maximum possible number of iterations of the **while** loop in the algorithm is _____.

Q.42 Consider the following context-free grammars:

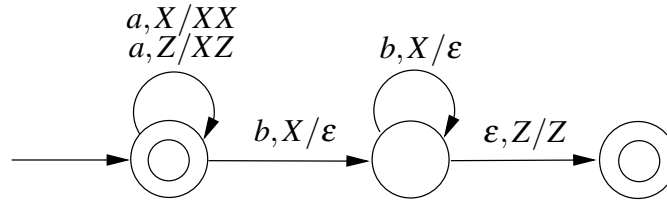
$G_1: S \rightarrow aS|B, B \rightarrow b|bB$

$G_2: S \rightarrow aA|bB, A \rightarrow aA|B|\epsilon, B \rightarrow bB|\epsilon$

Which one of the following pairs of languages is generated by G_1 and G_2 , respectively?

- (A) $\{a^m b^n | m > 0 \text{ or } n > 0\}$ and $\{a^m b^n | m > 0 \text{ and } n > 0\}$
- (B) $\{a^m b^n | m > 0 \text{ and } n > 0\}$ and $\{a^m b^n | m > 0 \text{ or } n \geq 0\}$
- (C) $\{a^m b^n | m \geq 0 \text{ or } n > 0\}$ and $\{a^m b^n | m > 0 \text{ and } n > 0\}$
- (D) $\{a^m b^n | m \geq 0 \text{ and } n > 0\}$ and $\{a^m b^n | m > 0 \text{ or } n > 0\}$

- Q.43** Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{X, Z\}$. Z is the initial stack symbol. Let L denote the language accepted by the PDA.



Which one of the following is **TRUE**?

- (A) $L = \{a^n b^n | n \geq 0\}$ and is not accepted by any finite automata
- (B) $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and is not accepted by any deterministic PDA
- (C) L is not accepted by any Turing machine that halts on every input
- (D) $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and is deterministic context-free
- Q.44** Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that $\bar{Y}$ reduces to W , and Z reduces to $\bar{X}$ (reduction means the standard many-one reduction). Which one of the following statements is **TRUE**?
- (A) W can be recursively enumerable and Z is recursive.
- (B) W can be recursive and Z is recursively enumerable.
- (C) W is not recursively enumerable and Z is recursive.
- (D) W is not recursively enumerable and Z is not recursive.

- Q.45** The attributes of three arithmetic operators in some programming language are given below.

Operator	Precedence	Associativity	Arity
+	High	Left	Binary
−	Medium	Right	Binary
*	Low	Left	Binary

The value of the expression $2 - 5 + 1 - 7 * 3$ in this language is _____.

Q.46 Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals $\{S, A\}$ and terminals $\{a, b\}$.

$S \rightarrow aA \quad \{ \text{print 1} \}$

$S \rightarrow a \quad \{ \text{print 2} \}$

$A \rightarrow Sb \quad \{ \text{print 3} \}$

Using the above SDTS, the output printed by a bottom-up parser, for the input **aab** is:

(A) 1 3 2

(B) 2 2 3

(C) 2 3 1

(D) syntax error

Q.47 Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is _____ megabytes.

Q.48 Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____.

Q.49 Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$, where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____.

- Q.50** Consider the following proposed solution for the critical section problem. There are n processes: $P_0 \dots P_{n-1}$. In the code, function `pmax` returns an integer not smaller than any of its arguments. For all i , $t[i]$ is initialized to zero.

Code for P_i :

```
do {
    c[i]=1; t[i] = pmax(t[0],...,t[n-1])+1; c[i]=0;
    for every j  $\neq$  i in {0,...,n-1} {
        while (c[j]);
        while (t[j] != 0 && t[j]<=t[i]);
    }
    Critical Section;
    t[i]=0;
    Remainder Section;
} while (true);
```

Which one of the following is **TRUE** about the above solution?

- (A) At most one process can be in the critical section at any time
- (B) The bounded wait condition is satisfied
- (C) The progress condition is satisfied
- (D) It cannot cause a deadlock

- Q.51** Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \dots, O_k\}$. This is done in the following manner:

Step 1. T acquires exclusive locks to $O_1, \dots, O_k$ in increasing order of their addresses.

Step 2. The required operations are performed.

Step 3. All locks are released.

This protocol will

- (A) guarantee serializability and deadlock-freedom
- (B) guarantee neither serializability nor deadlock-freedom
- (C) guarantee serializability but not deadlock-freedom
- (D) guarantee deadlock-freedom but not serializability

Q.52 Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted by K_x^- and K_x^+ for $x = A, B$, respectively. Let $K_x(m)$ represent the operation of encrypting m with a key K_x and $H(m)$ represent the message digest. Which one of the following indicates the **CORRECT** way of sending the message m along with the digital signature to A?

- (A) $\{m, K_B^+(H(m))\}$ (B) $\{m, K_B^-(H(m))\}$ (C) $\{m, K_A^-(H(m))\}$ (D) $\{m, K_A^+(m)\}$

Q.53 An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes.

The number of fragments that the IP datagram will be divided into for transmission is _____.

Q.54 For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

Q.55 A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds.

Assuming no frame is lost, the sender throughput is _____ bytes/second.

Q. No	Type	Section	Key	Marks
1	MCQ	GA	D	1
2	MCQ	GA	A	1
3	MCQ	GA	A	1
4	MCQ	GA	C	1
5	MCQ	GA	D	1
6	MCQ	GA	B	2
7	MCQ	GA	D	2
8	MCQ	GA	D	2
9	MCQ	GA	B	2
10	MCQ	GA	B	2
1	NAT	CS-1	11.0 : 11.0	1
2	MCQ	CS-1	B	1
3	NAT	CS-1	1.0 : 1.0	1
4	NAT	CS-1	0.5 : 0.5	1
5	NAT	CS-1	15.0 : 15.0	1
6	MCQ	CS-1	A	1
7	NAT	CS-1	-11.0 : -11.0	1
8	NAT	CS-1	3.0 : 3.0	1
9	NAT	CS-1	31.0 : 31.0	1
10	MCQ	CS-1	A	1
11	NAT	CS-1	6.0 : 6.0	1
12	MCQ	CS-1	D	1
13	MCQ	CS-1	D	1
14	MCQ	CS-1	A	1
15	NAT	CS-1	2016.0 : 2016.0	1
16	MCQ	CS-1	D	1
17	MCQ	CS-1	C	1
18	MCQ	CS-1	B	1
19	NAT	CS-1	10.0 : 10.0	1
20	MCQ	CS-1	A	1
21	MCQ	CS-1	B	1
22	MCQ	CS-1	D	1
23	MCQ	CS-1	B	1
24	MCQ	CS-1	C	1
25	MCQ	CS-1	C	1
26	NAT	CS-1	10.0 : 10.0	2
27	NAT	CS-1	197.9 : 198.1	2
28	NAT	CS-1	2.0 : 2.0	2
29	NAT	CS-1	0.33 : 0.34	2
30	MCQ	CS-1	D	2
31	NAT	CS-1	456.0 : 456.0	2
32	NAT	CS-1	33.0 : 34.0	2
33	MCQ	CS-1	B	2
34	MCQ	CS-1	D	2
35	MCQ	CS-1	A	2
36	MCQ	CS-1	D	2
37	MCQ	CS-1	B	2
38	NAT	CS-1	12.0 : 12.0	2
39	NAT	CS-1	7.0 : 7.0	2

40	MCQ	CS-1	B	2
41	NAT	CS-1	256.0 : 256.0	2
42	MCQ	CS-1	D	2
43	MCQ	CS-1	D	2
44	MCQ	CS-1	C	2
45	NAT	CS-1	9.0 : 9.0	2
46	MCQ	CS-1	C	2
47	NAT	CS-1	384.0 : 384.0	2
48	NAT	CS-1	346.0 : 346.0	2
49	NAT	CS-1	1.0 : 1.0	2
50	MCQ	CS-1	A	2
51	MCQ	CS-1	A	2
52	MCQ	CS-1	B	2
53	NAT	CS-1	13.0 : 13.0	2
54	NAT	CS-1	1.1 : 1.1	2
55	NAT	CS-1	2500.0 : 2500.0	2

Q. 1 – Q. 5 carry one mark each.

Q.1 The man who is now Municipal Commissioner worked as _____.

- (A) the security guard at a university
- (B) a security guard at the university
- (C) a security guard at university
- (D) the security guard at the university

Q.2 Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.

Choose the option which is closest in meaning to the underlined phrase in the above sentence.

- (A) put up with (B) put in with (C) put down to (D) put up against

Q.3 Find the odd one in the following group of words.

mock, deride, praise, jeer

- (A) mock (B) deride (C) praise (D) jeer

Q.4 Pick the odd one from the following options.

- (A) CADBE (B) JHKIL (C) XZYWZ (D) ONPMQ

Q.5 In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

- (A) n^4 (B) 4^n (C) 2^{2n-1} (D) 4^{n-1}

Q. 6 – Q. 10 carry two marks each.

Q.6 Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

- (A) 35 (B) 45 (C) 65 (D) 90

- Q.7 Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.

Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

- (i) The author believes that computers are not good for us.
- (ii) Mobile computers and the internet are both intended inventions

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.8 All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

- (i) Ooty is not a hill-station.
- (ii) No hill-station can have more than one lake.

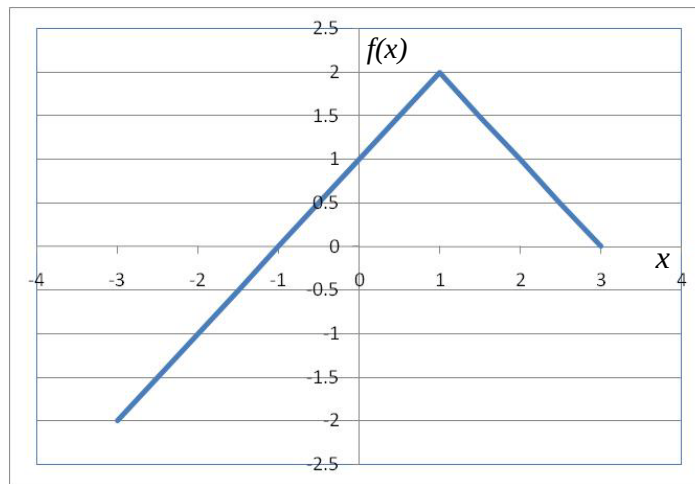
(A) (i) only (B) (ii) only
(C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.9 In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



(A) 21 (B) 27 (C) 30 (D) 36

Q.10



Choose the correct expression for $f(x)$ given in the graph.

(A) $f(x) = 1 - |x - 1|$

(B) $f(x) = 1 + |x - 1|$

(C) $f(x) = 2 - |x - 1|$

(D) $f(x) = 2 + |x - 1|$

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Consider the following expressions:

- (i) *false*
- (ii) Q
- (iii) *true*
- (iv) $P \vee Q$
- (v) $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____ .

Q.2 Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____ .

Q.3 The minimum number of colours that is sufficient to vertex-colour any planar graph is _____ .

Q.4 Consider the systems, each consisting of m linear equations in n variables.

- I. If $m < n$, then all such systems have a solution
- II. If $m > n$, then none of these systems has a solution
- III. If $m = n$, then there exists a system which has a solution

Which one of the following is **CORRECT**?

- (A) I, II and III are true
- (B) Only II and III are true
- (C) Only III is true
- (D) None of them is true

- Q.5** Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____ .
- Q.6** Suppose that the eigenvalues of matrix A are 1, 2, 4. The determinant of $(A^{-1})^T$ is _____ .
- Q.7** Consider an eight-bit ripple-carry adder for computing the sum of A and B , where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____ .
- Q.8** Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be **TRUE**?
- (A) $x_1 x_2 x_3 x_4 = 0$
 - (B) $x_1 x_3 + x_2 = 0$
 - (C) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$
 - (D) $x_1 + x_2 + x_3 + x_4 = 0$
- Q.9** Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X - Y$ is _____ .
- Q.10** A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____ .

Q.11 Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n -th vertex in this BFS traversal, then the maximum possible value of n is _____ .

Q.12 The value printed by the following program is _____ .

```
void f(int* p, int m){
    m = m + 5;
    *p = *p + m;
    return;
}

void main(){
    int i=5, j=10;

    f(&i, j);
    printf("%d", i+j);
}
```

Q.13 Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are **TRUE**?

- I. Quicksort runs in $\Theta(n^2)$ time
- II. Bubblesort runs in $\Theta(n^2)$ time
- III. Mergesort runs in $\Theta(n)$ time
- IV. Insertion sort runs in $\Theta(n)$ time

- (A) I and II only
- (B) I and III only
- (C) II and IV only
- (D) I and IV only

- Q.14** The Floyd-Warshall algorithm for all-pair shortest paths computation is based on
- (A) Greedy paradigm.
 - (B) Divide-and-Conquer paradigm.
 - (C) Dynamic Programming paradigm.
 - (D) neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.
- Q.15** N items are stored in a sorted doubly linked list. For a *delete* operation, a pointer is provided to the record to be deleted. For a *decrease-key* operation, a pointer is provided to the record on which the operation is to be performed.
- An algorithm performs the following operations on the list in this order: $\Theta(N)$ *delete*, $O(\log N)$ *insert*, $O(\log N)$ *find*, and $\Theta(N)$ *decrease-key*. What is the time complexity of all these operations put together?
- (A) $O(\log^2 N)$ (B) $O(N)$ (C) $O(N^2)$ (D) $\Theta(N^2 \log N)$
- Q.16** The number of states in the minimum sized DFA that accepts the language defined by the regular expression
- $$(0 + 1)^*(0 + 1)(0 + 1)^*$$
- is _____ .
- Q.17** Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b | \epsilon$
Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2 | \epsilon$
- Consider the following statements:
- P : L_1 is regular
 Q : L_2 is regular
- Which one of the following is **TRUE**?
- (A) Both P and Q are true
 - (B) P is true and Q is false
 - (C) P is false and Q is true
 - (D) Both P and Q are false

Q.18 Consider the following types of languages: L_1 : Regular, L_2 : Context-free, L_3 : Recursive, L_4 : Recursively enumerable. Which of the following is/are **TRUE**?

- I. $\overline{L_3} \cup L_4$ is recursively enumerable
- II. $\overline{L_2} \cup L_3$ is recursive
- III. $L_1^* \cap L_2$ is context-free
- IV. $L_1 \cup \overline{L_2}$ is context-free

- (A) I only
- (B) I and III only
- (C) I and IV only
- (D) I, II and III only

Q.19 Match the following:

- | | |
|--------------------------|---------------------------|
| (P) Lexical analysis | (i) Leftmost derivation |
| (Q) Top down parsing | (ii) Type checking |
| (R) Semantic analysis | (iii) Regular expressions |
| (S) Runtime environments | (iv) Activation records |

- (A) $P \leftrightarrow i, Q \leftrightarrow ii, R \leftrightarrow iv, S \leftrightarrow iii$
- (B) $P \leftrightarrow iii, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iv$
- (C) $P \leftrightarrow ii, Q \leftrightarrow iii, R \leftrightarrow i, S \leftrightarrow iv$
- (D) $P \leftrightarrow iv, Q \leftrightarrow i, R \leftrightarrow ii, S \leftrightarrow iii$

Q.20 In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

- (A) LRU (Least Recently Used)
- (B) OPT (Optimal Page Replacement)
- (C) MRU (Most Recently Used)
- (D) FIFO (First In First Out)

- Q.21** B+ Trees are considered **BALANCED** because
- (A) the lengths of the paths from the root to all leaf nodes are all equal.
 - (B) the lengths of the paths from the root to all leaf nodes differ from each other by at most 1.
 - (C) the number of children of any two non-leaf sibling nodes differ by at most 1.
 - (D) the number of records in any two leaf nodes differ by at most 1.
- Q.22** Suppose a database schedule S involves transactions $T_1, \dots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?
- (A) Topological order
 - (B) Depth-first order
 - (C) Breadth-first order
 - (D) Ascending order of transaction indices
- Q.23** Anarkali digitally signs a message and sends it to Salim. Verification of the signature by Salim requires
- (A) Anarkali's public key.
 - (B) Salim's public key.
 - (C) Salim's private key.
 - (D) Anarkali's private key.
- Q.24** In an Ethernet local area network, which one of the following statements is **TRUE**?
- (A) A station stops to sense the channel once it starts transmitting a frame.
 - (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
 - (C) A station continues to transmit the packet even after the collision is detected.
 - (D) The exponential backoff mechanism reduces the probability of collision on retransmissions.

Q.25 Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

- (A) HTTP GET request, DNS query, TCP SYN
- (B) DNS query, HTTP GET request, TCP SYN
- (C) DNS query, TCP SYN, HTTP GET request
- (D) TCP SYN, DNS query, HTTP GET request

Q. 26 - Q. 55 carry two marks each.

Q.26 A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a, b)R(c, d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

P: R is reflexive

Q: R is transitive

Which one of the following statements is **TRUE**?

- (A) Both P and Q are true.
- (B) P is true and Q is false.
- (C) P is false and Q is true.
- (D) Both P and Q are false.

Q.27 Which one of the following well-formed formulae in predicate calculus is **NOT** valid?

- (A) $(\forall x p(x) \Rightarrow \forall x q(x)) \Rightarrow (\exists x \neg p(x) \vee \forall x q(x))$
- (B) $(\exists x p(x) \vee \exists x q(x)) \Rightarrow \exists x (p(x) \vee q(x))$
- (C) $\exists x (p(x) \wedge q(x)) \Rightarrow (\exists x p(x) \wedge \exists x q(x))$
- (D) $\forall x (p(x) \vee q(x)) \Rightarrow (\forall x p(x) \vee \forall x q(x))$

Q.28 Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

- (A) Only I
- (B) Only II
- (C) Only III
- (D) None

Q.29 The value of the expression $13^{99} \pmod{17}$, in the range 0 to 16, is _____.

Q.30 Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G , respectively. Given two instances of U_F and two instances of U_G , it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____ nanoseconds.

Q.31 Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _____.

Q.32 The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is _____ bits.

- Q.33** Consider a 3 GHz (gigahertz) processor with a three-stage pipeline and stage latencies τ_1 , τ_2 , and τ_3 such that $\tau_1 = 3\tau_2/4 = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.
- Q.34** A complete binary min-heap is made by including each integer in $[1, 1023]$ exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is _____.
- Q.35** The following function computes X^Y for positive integers X and Y.

```
int exp(int X, int Y) {  
    int res = 1, a = X, b = Y;  
  
    while ( b != 0 ){  
        if ( b%2 == 0 ) { a = a*a; b = b/2; }  
        else { res = res*a; b = b-1; }  
    }  
    return res;  
}
```

Which one of the following conditions is **TRUE** before every iteration of the loop?

- (A) $X^Y = a^b$
- (B) $(res * a)^Y = (res * X)^b$
- (C) $X^Y = res * a^b$
- (D) $X^Y = (res * a)^b$

Q.36 Consider the following New-order strategy for traversing a binary tree:

- Visit the root;
- Visit the right subtree using New-order;
- Visit the left subtree using New-order;

The New-order traversal of the expression tree corresponding to the reverse polish expression $3\ 4\ *\ 5\ -\ 2\ ^\ 6\ 7\ *\ 1\ +\ -$ is given by:

- (A) $+ - 1\ 6\ 7\ *\ 2\ ^\ 5 - 3\ 4\ *$
(B) $- + 1\ *\ 6\ 7\ ^\ 2 - 5\ *\ 3\ 4$
(C) $- + 1\ *\ 7\ 6\ ^\ 2 - 5\ *\ 4\ 3$
(D) $1\ 7\ 6\ *\ +\ 2\ 5\ 4\ 3\ *\ -\ ^\ -$

Q.37 Consider the following program:

```
int f(int *p, int n)
{
    if (n <= 1) return 0;
    else return max(f(p+1,n-1),p[0]-p[1]);
}

int main()
{
    int a[] = {3,5,2,6,4};
    printf("%d", f(a,5));
}
```

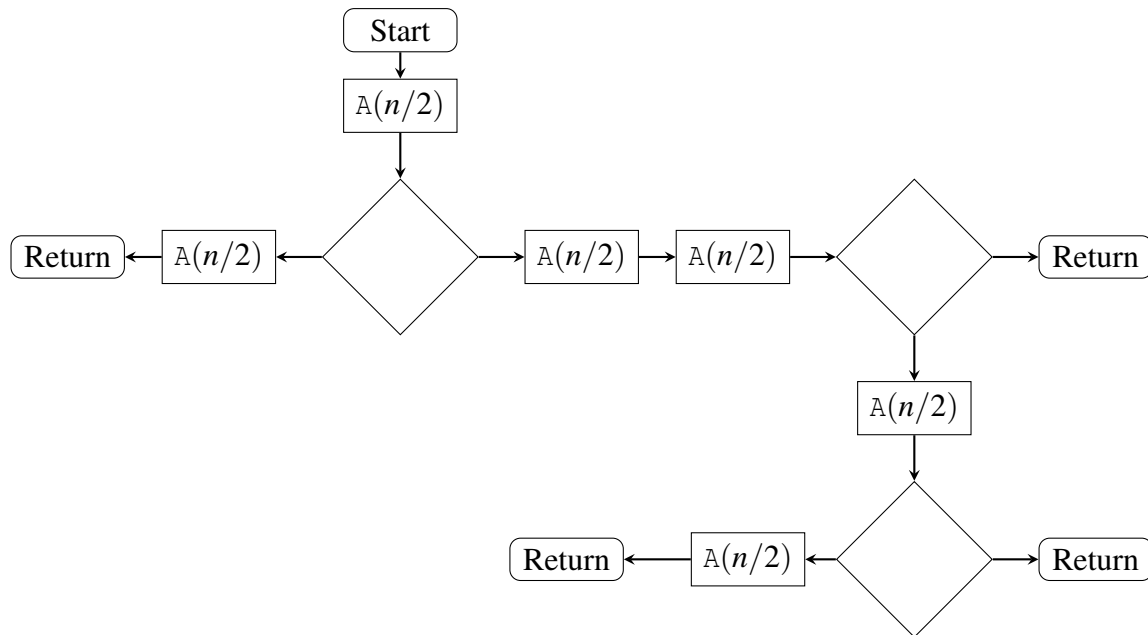
Note: $\max(x, y)$ returns the maximum of x and y .

The value printed by this program is _____ .

Q.38 Let A_1, A_2, A_3 , and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _____ .

- Q.39** The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is _____.

Flowchart for Recursive Function $A(n)$



- Q.40** The number of ways in which the numbers 1, 2, 3, 4, 5, 6, 7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is _____ .
- Note: The height of a tree with a single node is 0.*

Q.41 In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u , and $[u]$ in the adjacency list of v . These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E| = m$ and $|V| = n$, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

- (A) $\Theta(n^2)$
- (B) $\Theta(n + m)$
- (C) $\Theta(m^2)$
- (D) $\Theta(n^4)$

Q.42 Consider the following two statements:

- I.** If all states of an NFA are accepting states then the language accepted by the NFA is Σ^* .
- II.** There exists a regular language A such that for all languages B , $A \cap B$ is regular.

Which one of the following is **CORRECT**?

- (A) Only **I** is true
- (B) Only **II** is true
- (C) Both **I** and **II** are true
- (D) Both **I** and **II** are false

Q.43 Consider the following languages:

$$L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$$

$$L_2 = \{a^n b^n c^{2n} : n \geq 1\}$$

Which one of the following is **TRUE**?

- (A) Both L_1 and L_2 are context-free.
- (B) L_1 is context-free while L_2 is not context-free.
- (C) L_2 is context-free while L_1 is not context-free.
- (D) Neither L_1 nor L_2 is context-free.

Q.44 Consider the following languages.

$$\begin{aligned} L_1 &= \{ \langle M \rangle \mid M \text{ takes at least 2016 steps on some input} \}, \\ L_2 &= \{ \langle M \rangle \mid M \text{ takes at least 2016 steps on all inputs} \} \text{ and} \\ L_3 &= \{ \langle M \rangle \mid M \text{ accepts } \varepsilon \}, \end{aligned}$$

where for each Turing machine M , $\langle M \rangle$ denotes a specific encoding of M . Which one of the following is **TRUE**?

- (A) L_1 is recursive and L_2, L_3 are not recursive
- (B) L_2 is recursive and L_1, L_3 are not recursive
- (C) L_1, L_2 are recursive and L_3 is not recursive
- (D) L_1, L_2, L_3 are recursive

Q.45 Which one of the following grammars is free from *left recursion*?

- (A)
$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow Aa \mid b \\ B &\rightarrow c \end{aligned}$$
- (B)
$$\begin{aligned} S &\rightarrow Ab \mid Bb \mid c \\ A &\rightarrow Bd \mid \varepsilon \\ B &\rightarrow e \end{aligned}$$
- (C)
$$\begin{aligned} S &\rightarrow Aa \mid B \\ A &\rightarrow Bb \mid Sc \mid \varepsilon \\ B &\rightarrow d \end{aligned}$$
- (D)
$$\begin{aligned} S &\rightarrow Aa \mid Bb \mid c \\ A &\rightarrow Bd \mid \varepsilon \\ B &\rightarrow Ae \mid \varepsilon \end{aligned}$$

- Q.46** A student wrote two context-free grammars **G1** and **G2** for generating a single C-like array declaration. The dimension of the array is at least one. For example,

```
int a[10][3];
```

The grammars use D as the start symbol, and use six terminal symbols **int ; id [] num**.

Grammar **G1**

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}[E$

$E \rightarrow \text{num}]$

$E \rightarrow \text{num}][E$

Grammar **G2**

$D \rightarrow \text{int}L;$

$L \rightarrow \text{id}E$

$E \rightarrow E[\text{num}]$

$E \rightarrow [\text{num}]$

Which of the grammars correctly generate the declaration mentioned above?

- (A) Both **G1** and **G2**
- (B) Only **G1**
- (C) Only **G2**
- (D) Neither **G1** nor **G2**

- Q.47** Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes is _____ milliseconds.

Q.48 Consider the following two-process synchronization solution.

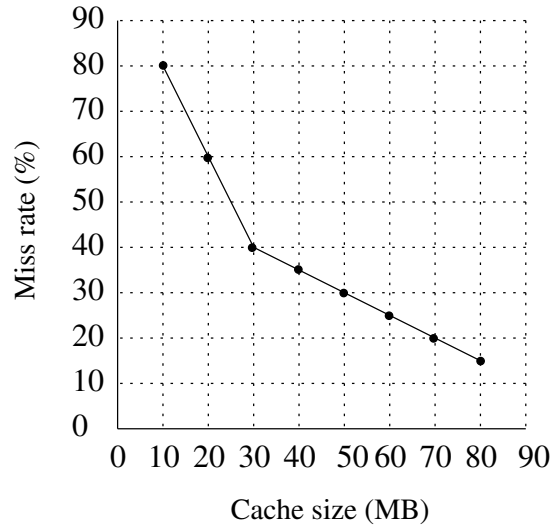
Process 0	Process 1
-----	-----
Entry: loop while (turn == 1);	Entry: loop while (turn == 0);
(critical section)	(critical section)
Exit: turn = 1;	Exit: turn = 0;

The shared variable `turn` is initialized to zero. Which one of the following is **TRUE**?

- (A) This is a correct two-process synchronization solution.
- (B) This solution violates mutual exclusion requirement.
- (C) This solution violates progress requirement.
- (D) This solution violates bounded wait requirement.

Q.49 Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is _____.

- Q.50** A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

- Q.51** Consider the following database schedule with two transactions, T_1 and T_2 .

$$S = r_2(X); r_1(X); r_2(Y); w_1(X); r_1(Y); w_2(X); a_1; a_2$$

where $r_i(Z)$ denotes a *read* operation by transaction T_i on a variable Z , $w_i(Z)$ denotes a *write* operation by T_i on a variable Z and a_i denotes an *abort* by transaction T_i .

Which one of the following statements about the above schedule is **TRUE**?

- (A) S is non-recoverable
- (B) S is recoverable, but has a cascading abort
- (C) S does not have a cascading abort
- (D) S is strict

Q.52 Consider the following database table named *water_schemes* :

<i>water_schemes</i>		
scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____ .

```
with total(name, capacity) as
  select district_name, sum(capacity)
  from water_schemes
  group by district_name
with total_avg(capacity) as
  select avg(capacity)
  from total
select name
  from total, total_avg
  where total.capacity ≥ total_avg.capacity
```

Q.53 A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

Q.54 For the IEEE 802.11 MAC protocol for wireless communication, which of the following statements is/are **TRUE**?

- I. At least three non-overlapping channels are available for transmissions.
- II. The RTS-CTS mechanism is used for collision detection.
- III. Unicast frames are ACKed.

- (A) All I, II, and III
- (B) I and III only
- (C) II and III only
- (D) II only

Q.55 Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____ .

Q. No	Type	Section	Key	Marks
1	MCQ	GA	B	1
2	MCQ	GA	A	1
3	MCQ	GA	C	1
4	MCQ	GA	D	1
5	MCQ	GA	B	1
6	MCQ	GA	A	2
7	MCQ	GA	D	2
8	MCQ	GA	D	2
9	MCQ	GA	C	2
10	MCQ	GA	C	2
1	NAT	CS-2	4.0 : 4.0	1
2	NAT	CS-2	9.0 : 9.0	1
3	NAT	CS-2	4.0 : 4.0	1
4	MCQ	CS-2	C	1
5	NAT	CS-2	0.55 : 0.55	1
6	NAT	CS-2	0.124 : 0.126	1
7	NAT	CS-2	-1.0 : -1.0	1
8	MCQ	CS-2	C	1
9	NAT	CS-2	1.0 : 1.0	1
10	NAT	CS-2	16.0 : 16.0	1
11	NAT	CS-2	31.0 : 31.0	1
12	NAT	CS-2	30.0 : 30.0	1
13	MCQ	CS-2	D	1
14	MCQ	CS-2	C	1
15	MCQ	CS-2	C	1
16	NAT	CS-2	2.0 : 2.0	1
17	MCQ	CS-2	C	1
18	MCQ	CS-2	D	1
19	MCQ	CS-2	B	1
20	MCQ	CS-2	D	1
21	MCQ	CS-2	A	1
22	MCQ	CS-2	A	1
23	MCQ	CS-2	A	1
24	MCQ	CS-2	D	1
25	MCQ	CS-2	C	1
26	MCQ	CS-2	B	2
27	MCQ	CS-2	D	2
28	MCQ	CS-2	B	2
29	NAT	CS-2	4.0 : 4.0	2
30	NAT	CS-2	28.0 : 28.0	2
31	NAT	CS-2	500.0 : 500.0	2
32	NAT	CS-2	24.0 : 24.0	2
33	NAT	CS-2	4.0 : 4.0	2
34	NAT	CS-2	8.0 : 8.0	2
35	MCQ	CS-2	C	2
36	MCQ	CS-2	C	2
37	NAT	CS-2	3.0 : 3.0	2
38	NAT	CS-2	1500.0 : 1500.0	2
39	NAT	CS-2	2.2 : 2.4	2

40	NAT	CS-2	64.0 : 64.0	2
41	MCQ	CS-2	B	2
42	MCQ	CS-2	B	2
43	MCQ	CS-2	B	2
44	MCQ	CS-2	C	2
45	MCQ	CS-2	B	2
46	MCQ	CS-2	A	2
47	NAT	CS-2	8.2 : 8.3	2
48	MCQ	CS-2	C	2
49	NAT	CS-2	7.0 : 7.0	2
50	NAT	CS-2	30.0 : 30.0	2
51	MCQ	CS-2	C	2
52	NAT	CS-2	2.0 : 2.0	2
53	NAT	CS-2	200.0 : 200.0	2
54	MCQ	CS-2	B	2
55	NAT	CS-2	4.0 : 4.0	2